

A
PROJECT REPORT
ON
“INTERNSHIP PORTAL”
IN PARTIAL FULFILLMENT OF
MASTER OF COMPUTER APPLICATION
BY
SHRUTI CHANDRAKANT PAWAR
MCA-II SEM-IV
UNDER THE GUIDANCE OF
PROF. PRAVIN BAGAL
SUBMITTED TO
SAVITRIBAI PHULE PUNE UNIVERSITY
(2025-2026)

VIDYA PRATISHTHAN’S
INSTITUTE OF INFORMATION TECHNOLOGY (VIIT),
(ACCREDITED BY NAAC WITH “B” GRADE, CYCLE-II)
VIDYA NAGARI, BARAMATI,
DIST. PUNE 413 133



Vidya Pratishthan's
Institute of Information Technology
We Shape Tomorrow...Today



NAAC
NATIONAL ASSESSMENT AND
ACCREDITATION COUNCIL

"B"
Grade
CYCLE II

CERTIFICATE

This is to certify that Miss. Shruti Pawar of MCA Course, SEM-IV have completed her Project work Titled “**Internship Portal**” as a part of the curriculum, during the Academic Year 2025-26.

Dr. Ajit Adsul

HOD, MCA

Dr. Anand Deshmukh

Director, VIIT

External Examiners:

1. _____

2. _____

Sign of Examiners:

1. _____

2. _____



SANYU INFOTECH PVT LTD

Internship Completion Certificate

Date: 13/05/2026
Reg. No. SIPL/189/2026

Sanyu Infotech Pvt Ltd.
Karve Nagar, Pune. 411052.

Subject: Internship Completion Letter.

To Whom It May Concern,

This is to certify that **Shruti chandrakant pawar** from **Vidya Pratisthan's Institute of Information Technology (VIIT), Baramati** has successfully completed an internship at **Sanyu Infotech Pvt. Ltd.** from **02 Jan 2026 to 13 May 2026** as a **Full Stack Java Intern**.

During this internship, the candidate worked on real time projects and demonstrated strong technical skills, dedication, and a professional approach toward assigned responsibilities.

We appreciate the efforts and contributions made during the internship period and wish them continued success in their future career.

Yours Sincerely,
Sanyu Infotech Pvt. Ltd.


Senior HR
(Nayan Dhulgand)



 Sanyu Infotech Pvt Ltd,
Vikas Mitra Mandal Chowk,
Karve Nagar, Pune.
Pin Code-411052

 hr@sanyuinfotech.com

 91 3030 5960

 www.sanyuinfotech.com





Vidya Pratishthan's
Institute of Information Technology
We Shape Tomorrow...Today



NAAC
NATIONAL ASSESSMENT AND
ACCREDITATION COUNCIL

"B"
Grade
CYCLE II

GUIDE CERTIFICATE

This is to certify that Miss. Shruti Pawar of MCA Course, SEM- IV have successfully completed her Project Work Title “**Internship Portal**”, under my guidance during the Academic Year 2025-26.

Prof. Pravin Bagal

Project Guide



Vidya Pratishthan's
Institute of Information Technology
We Shape Tomorrow...Today



NAAC
NATIONAL ASSESSMENT AND
ACCREDITATION COUNCIL

"B"
Grade
CYCLE II

DECLARATION

I hereby declare that, the project entitled "**Internship Portal**" have completed and written by me. Current project has not previously formed the basis for the award of any Degree or Diploma or other similar title of this or any other University or examining body.

Place: Baramati

Date:

Signature

SHRUTI PAWAR



Vidya Pratishthan's
Institute of Information Technology
We Shape Tomorrow...Today



NAAC
NATIONAL ASSESSMENT AND
ACCREDITATION COUNCIL

"B"
Grade
CYCLE II

ACKNOWLEDGEMENT

I take pleasure in presenting our project work. I am thankful to our project guide **Prof. Pravin Bagal** of VIIT, who keeps monitoring our project continuously to complete it in time as per the expectations of the course.

"Ability is of little account without opportunity", I wish to thank our Director **Dr. Anand Deshmukh**, who gave me a very bright learning opportunity, during our course to prove our ability and skills.

I would also like to thank **Dr. Ajit Adsul**, Head of Department, MCA for providing all the necessary facilities at college during the development of this project.

Last but not least I would like to express our heartfelt gratitude towards staff members of VIIT, our colleagues and friends for their moral and technical support throughout the duration of the project.

Our sincere thanks to all.

SHRUTI PAWAR

Plagiarism Report



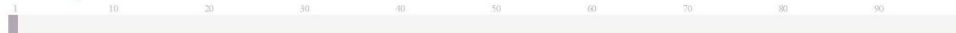
The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

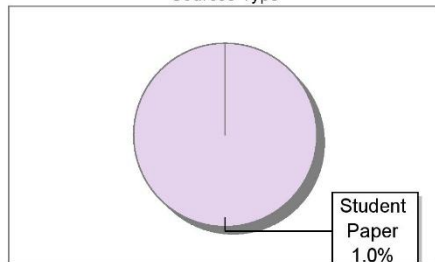
Author Name	Shruti Pawar, Pooja Kale
Title	Internship Portal
Paper/Submission ID	5869866
Submitted by	pravin.bagal@viitindia.org
Submission Date	2026-06-05 15:10:24
Total Pages, Total Words	64, 7261
Document type	Project Work

Result Information

Similarity **1 %**



Sources Type



Exclude Information

Quotes	Not Excluded
References/Bibliography	Not Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	0 %
Excluded Phrases	Not Excluded

Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File



TABLE OF CONTENTS

Chapter No.	Details	Page No.
1	Introduction	01
1.1	Company Profile	01
1.2	Problem Statement	03
1.3	Existing System and Need for the System	04
1.4	Scope of the Proposed System	05
1.5	Objectives of the Proposed System	07
1.6	Operating Environment - Hardware and Software	07
1.7	Brief Description of Technology Used for Frontend, Backend & Database.	09
2	System Analysis & Design	11
2.1	System Requirements (Functional and Non-Functional requirements)	11
2.2	Feasibility Study (Technical, Economic & Operational)	15
2.3	Architecture Design	17
2.4	Entity Relationship Diagram (ERD)	20
2.5	Class Diagram	21
2.6	Use Case Diagrams	22
2.7	Activity Diagram	25
2.8	Sequence Diagram	26
2.9	Component Diagram	27
2.10	Module Hierarchy Diagram	28
2.11	Table Design	29
2.12	Data Dictionary	32
3	Implementation	35
3.1	Input Screens	35
3.2	Output Screens/Reports	39
3.3	Sample code	44
4	Testing	49
4.1	Test Strategy	49
4.2	Unit Test Plan	49
4.3	Acceptance of Test Plan	50
4.4	Test Cases	52
4.5	Results	54
5	Conclusion	55
5.1	Summary/Conclusion	55
5.2	Limitations of the System	56
5.3	Future Enhancements	57
6	References / Bibliography	58
7	Appendices	61
7.1	Annexure I- Additional Input & Output Screens	61
7.2	Annexure II- Annexure -Progress Sheet	64

Chapter 1: Introduction

1.1 Company Profile – Sanyu Infotech Pvt. Ltd.

Sanyu Infotech Pvt. Ltd. is a Pune-based IT company that works in the fields of software development, digital technologies, IT training, and internship programs. The company is dedicated to providing modern and practical technology solutions for businesses while also helping students gain real industry knowledge and hands-on experience.

One of the main goals of the company is to bridge the gap between academic education and industry expectations. Through live projects, practical training, and expert guidance, Sanyu Infotech Pvt. Ltd. helps students understand real working environments in the IT sector and prepares them for future career opportunities.

Founding and Location

Sanyu Infotech Pvt. Ltd. is located near Vikas Mitra Mandal Chowk, Karve Nagar, Pune – 411052. The company was established with the aim of delivering innovative software solutions and creating career opportunities for students and fresh graduates interested in the technology field. Over the years, the organization has grown its services in software development, digital transformation, and technical education.

Vision and Mission

The vision of Sanyu Infotech Pvt. Ltd. is to empower businesses and individuals through advanced digital technologies and smart software solutions. The company believes that technology can play an important role in solving real-world challenges and improving business efficiency.

The mission of the organization is to convert innovative ideas into practical IT solutions while helping students and professionals develop technical skills and industry exposure. The company encourages values like teamwork, innovation, transparency, and continuous improvement.

Services and Training Programs

Sanyu Infotech Pvt. Ltd. provides a wide range of IT services and training programs in different modern technologies, including:

- Software Development
- Full Stack Development
- Java and MERN Stack Technologies
- Cloud Computing
- DevOps Solutions
- Testing and Quality Analysis
- Data Analytics
- Artificial Intelligence and Machine Learning

Apart from technical services, the company also conducts internship programs, live project training, workshops, and placement preparation sessions. These programs are designed to improve students' practical knowledge and professional skills.

Work Culture and Industry Approach

Sanyu Infotech Pvt. Ltd. follows a practical, learning-oriented, and innovation-driven work culture. The company focuses on building scalable and efficient digital solutions that help businesses improve productivity and performance. By using modern technologies and real-time project development, the organization ensures quality service delivery.

The company also believes strongly in practical learning and mentorship. Students working with the organization get opportunities to participate in live projects, teamwork activities, and guided learning experiences, which help them gain confidence and understand industry practices better.

Internship and Career Development

Sanyu Infotech Pvt. Ltd. gives special importance to student career development. The company offers internship opportunities where students and freshers can work on real-time projects and applications. During the internship period, students improve their technical knowledge, communication skills, teamwork abilities, and problem-solving skills.

The guidance and training provided by the company help students become industry-ready and improve their chances of securing good career opportunities in the IT sector.

1.2 Problem Statement :-

The current internship management process is unorganized, time-consuming, and inefficient for students, companies, and administrators. Most internship activities are handled manually through emails, spreadsheets, social media platforms, and different job portals. Because there is no centralized system, the complete process becomes scattered and difficult to manage properly.

Students often face problems while searching for suitable internship opportunities. They need to check multiple websites, emails, and notices regularly to find openings. Many students miss good opportunities because information is not available in one place. Applying for internships is also repetitive, as students must fill different forms and upload resumes again and again for each company.

Another major issue is the lack of transparency in the current system. After applying, students usually do not receive proper updates about their application status. They remain confused about whether they are shortlisted, rejected, selected, or still under review. This creates uncertainty and stress. Companies also face several challenges in the manual system. They receive many applications through emails, Google Forms, or external portals. Managing and sorting large numbers of resumes manually takes significant time and effort. Recruiters need to review applications one by one, maintain records in spreadsheets, and contact candidates separately.

Manual record management increases the chances of duplicate entries, missing information, delayed responses, and human errors. Communication gaps frequently occur between students and companies because there is no structured platform for interaction.

Administrators or placement officers also face difficulty in maintaining student records, coordinating with companies, and generating reports. Preparing statistics related to applications, selections becomes slow and complicated. The current system is not suitable for handling a growing number of students and internship opportunities efficiently. It lacks automation, proper tracking, transparency, and secure data management.

1.3 Existing System and Need for the System-:

Existing System

The current internship management process is mostly handled manually through emails, WhatsApp groups, notice boards, spreadsheets, and external job portals. Because there is no centralized platform, the process becomes time-consuming and difficult to manage.

Problems in the Existing System:

- Students need to search multiple websites and platforms for internship opportunities.
- Important updates are often missed because information is scattered.
- Students repeatedly upload resumes and fill different application forms.
- There is no proper system to track application status.
- Communication between students and companies is slow and unorganized.
- Companies manually manage resumes and applications through emails or spreadsheets.
- Manual record handling increases chances of duplicate data and human errors.
- Administrators face difficulty in maintaining records and generating reports.
- The overall process lacks transparency, automation, and efficiency.

Need for the System

To solve these problems, there is a need for a centralized and automated Internship Portal that connects students, companies, and administrators on one platform.

The proposed system is needed to:

- Provide a single platform for all internship-related activities.
- Help students easily search and apply for internships.
- Reduce manual paperwork and repetitive tasks.
- Allow students to track application status in real time.
- Help companies manage applications efficiently.
- Improve communication between students and recruiters.

- Maintain secure and organized data storage.
- Reduce administrative workload and human errors.
- Generate reports and internship statistics easily.
- Create a transparent, scalable, and user-friendly system.

The Internship Portal modernizes the traditional internship process by making it faster, more organized, and more efficient for all users.

1.4 Scope of the Proposed System-:

The Internship Portal is designed to provide a complete and centralized digital platform for managing internship-related activities efficiently. The system connects students, companies, and administrators through a single web-based application, reducing manual work and improving communication.

The scope of the system includes the complete internship lifecycle starting from registration to final selection. Students can create profiles, upload resumes, search internship opportunities, apply online, and track their application status. Companies can register, create company profiles, post internship openings, review applications, download resumes, and update candidate selection status. Administrators can monitor the overall platform, manage users, approve internship postings, and maintain system records.

The system also includes communication and support features such as contact form inquiries, admin replies, direct messaging, and notification alerts. Users receive updates about application progress, messages, approvals, and important system activities through the notification module.

The platform supports secure login, role-based access control, centralized database management, and organized data storage. It provides dashboards and reports for monitoring system performance, user growth, and internship statistics.

The Internship Portal is developed as a responsive web application, allowing users to access the system through desktops, laptops, tablets, or mobile browsers using an internet connection.

What is Included in the Scope?

- Student registration and profile management
- Resume upload and profile updates

- Internship browsing, searching, and filtering
- Online internship application submission
- Application status tracking
- Company registration and profile management
- Internship posting, editing, and deletion
- Applicant review and resume download
- Candidate selection / rejection process
- Admin management of users and internships
- Contact form and support messaging
- Direct admin messaging
- Real-time notification system
- Dashboard and report generation
- Public browsing of internships

What is Not Included in the Current Scope?

- Payment gateway or paid subscriptions
- Social media login integration
- Dedicated Android or iOS mobile application
- Built-in video interviews or meeting tools
- AI-based internship recommendations
- Automated email or SMS notifications
- Bulk status update operations
- Student-company live chat system
- Two-factor authentication
- Advanced analytics and export tools
- Company rating and review system
- Internship bookmarking or wishlist
- Interview scheduling calendar

Overall, the scope of the Internship Portal is to modernize the traditional internship process by replacing scattered manual methods with a secure, scalable, transparent, and user-friendly online system that benefits students, companies, and administrators.

1.5 Objectives of Proposed System -:

1. To develop a centralized platform for internship management.
2. To reduce manual work and paperwork.
3. To provide role-based access for students, companies, and admin.
4. To allow students to easily search and apply for internships.
5. To help companies manage applications efficiently.
6. To provide real-time tracking of application status.
7. To improve communication between students and companies.
8. To maintain secure and organized data storage.
9. To generate reports easily for monitoring purposes.
10. To create a scalable and user-friendly system.

1.6 Operating Environment - Hardware and Software -:

Operational requirements define the minimum hardware and software environment required for the Internship Management Platform to operate efficiently on both client and server sides.

1.6.1 Client-Side Requirements:

Hardware:

- Minimum Intel i3 processor or equivalent
- At least 4 GB RAM
- 20 GB free storage
- Stable internet connection

Software:

- Operating System: Windows, Linux, or macOS
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge

1.5.1: Server-Side Requirements:

Hardware :

- Minimum Intel i5 processor or higher
- At least 8 GB RAM
- Minimum 100 GB storage
- High-speed internet connectivity

Software :

- Operating System: Windows Server or Linux
- Java JDK 17 or higher
- Spring Boot Framework
- Apache Maven (for dependency management)
- Tomcat Server

1.7 Brief Description of Technology Used for Frontend, Backend & Database-:

The Internship Portal is developed using modern web technologies to provide a secure, responsive, scalable, and user-friendly platform. The system follows a frontend-backend architecture where different technologies work together to manage user interaction, business logic, and database operations efficiently.

Frontend Technologies -:

- **React.js**

React.js is a JavaScript library used for building the user interface of the application. It helps create reusable components and provides a fast, responsive, and interactive user experience. React allows smooth page navigation without reloading the entire webpage.

- **HTML5**

HTML5 is used to create the basic structure of web pages. It defines elements such as forms, buttons, tables, input fields, and layouts used throughout the Internship Portal.

- **CSS3**

CSS3 is used for styling and designing the application interface. It improves the visual appearance of the system by adding colors, layouts, spacing, responsiveness, and animations.

- **JavaScript**

JavaScript is used to add dynamic functionality and interactivity to the application. It handles events, validations, form interactions, and frontend logic.

- **Axios**

Axios is a JavaScript library used for making HTTP requests between the frontend and backend. It helps send and receive data from REST APIs efficiently.

Backend Technologies -:

- **Java**

Java is the core programming language used for backend development. It is secure, platform-independent, and suitable for building scalable enterprise applications.

- **Spring Boot**

Spring Boot is a Java framework used to develop RESTful APIs and handle server-side

business logic.

- **Maven**

Maven is a build automation and dependency management tool used in the project. It manages libraries, plugins, and project configurations efficiently.

- **Tomcat Server**

Apache Tomcat is used as the application server to run and deploy the Spring Boot application. It handles client requests and server responses.

Database Technology

- **MySQL**

MySQL is a relational database management system used to store and manage system data such as users, internships, applications, notifications, and company details securely. It supports efficient data retrieval and structured storage.

Chapter 2: System Analysis and Design

2.1 System Requirements (Functional and Non-Functional Requirements)

2.1.1 Functional Requirements

Functional requirements define the main features and operations of the Internship Portal system for Students, Companies, and Admin users.

1. User Authentication Module

- Student and company registration with required details
- Secure login using email and password
- Role-based access for Admin, Student, and Company
- Secure logout functionality
- Prevention of unauthorized access

2. Student Module

- Create and update student profile
- Upload resume in PDF/document format
- Add academic details, skills, and projects
- Browse and search internships
- Apply for internships
- Track application status
- Receive notifications and updates

3. Company Module

- Create and manage company profile
- Post, edit, and delete internship opportunities

- View student applications and resumes
- Shortlist, reject, or select candidates
- Receive notifications for new application

4. Admin Module

- Secure admin login
- View registered students and companies
- Approve or reject company registrations and internships
- Remove invalid or inactive users
- Monitor system activities
- Send messages and generate reports

5. Internship Module

- Create and manage internship postings
- Include details like title, stipend, location, skills, and deadline
- Allow students to view and search internships
- Enable public users to browse internships

6. Internship Application Module

- Apply for internships through application forms
- Upload resumes during application
- Save application date and time
- Allow companies to review applications
- Update application status such as Applied, Shortlisted, Rejected, or Selected

7. Contact Module

- Submit contact forms with name, email, and message
- Validate required fields before submission

- Store messages in the database
- Allow admin to view and reply to messages
- Enable users to view admin replies

8. Notification Module

- Generate notifications automatically for important events
- Display unread notification count
- Allow users to view and mark notifications as read
- Redirect users to related pages through notifications
- Store notification history

9. Report Module

- View total students, companies, internships, and applications
- Generate reports for selected and rejected candidates
- Display active internship statistics

2.1.2 Non-Functional Requirements

Non-functional requirements define the quality, performance, and reliability of the Internship Portal system.

1. Performance Requirements

- Fast page loading and response time
- Support multiple users simultaneously
- Quick processing of applications and notifications
- Efficient database operations

2. Security Requirements

- Secure login with encrypted passwords

- Role-based access control
- Protection of user data and resumes
- Prevention of unauthorized database access and SQL Injection

3. Reliability Requirements

- Smooth system operation with minimal failures
- Accurate and consistent data management
- Proper error handling and validation
- Data recovery during unexpected failures

4. Usability Requirements

- Simple and user-friendly interface
- Easy navigation for all users
- Responsive design for mobile and desktop devices
- Clear notifications and status updates

5. Scalability Requirements

- Support increasing users and internship data
- Easy addition of future features and modules
- Efficient handling of large database records

6. Availability Requirements

- Accessible anytime through internet connection
- Minimal downtime during usage
- Continuous server availability

7. Maintainability Requirements

- Easy system updates and maintenance
- Modular and organized code structure

- Simple implementation of bug fixes and enhancements

8. Compatibility Requirements

- Support for Windows, Linux, and macOS
- Compatibility with browsers like Chrome, Firefox, and Edge

9. Backup and Recovery Requirements

- Regular database backups
- Data recovery during system failure
- Secure and reliable backup process

10. Portability Requirements

- Deployable on different server environments
- Support future migration to cloud platforms

2.2 Feasibility Study -:

A feasibility study is conducted to evaluate whether the proposed Internship Portal system is practical, cost- effective, and beneficial for users. It helps in understanding the viability of the system before development and implementation. The feasibility of the system is analyzed under three main categories:

2.2.1 Technical Feasibility

Technical feasibility determines whether the required technology, tools, and resources are available to develop and run the system successfully.

The Internship Portal is technically feasible because it is developed using widely used and reliable technologies such as React.js, Spring Boot, Java, and MySQL. These technologies are well-supported, scalable, and suitable for building modern web applications.

The system does not require any specialized or expensive hardware. It can run on standard computers with basic configurations such as a minimum of 4GB RAM and a stable internet connection. On the server side, commonly available environments like Apache Tomcat, Java JDK, and Maven are used, which are easy to install and configure.

The development tools required for the project, such as VS Code, Eclipse, and MySQL

Workbench, are freely available. The technologies used also support integration and future enhancements like mobile applications and AI-based features.

Since the developers are familiar with these technologies, the system can be developed, tested, and maintained efficiently. Therefore, the project is technically feasible.

2.2.2 Economic Feasibility

Economic feasibility evaluates the cost required to develop and maintain the system compared to the expected benefits.

The Internship Portal is economically feasible because it requires minimal investment. The technologies used in the project are open-source and free to use, which reduces development cost significantly. There is no need to purchase expensive licenses or software.

The system reduces operational costs by eliminating manual work such as paperwork, printing, and physical record maintenance. It also reduces administrative effort by automating tasks like application tracking, data storage, and report generation.

For companies, the system reduces recruitment costs by providing a centralized platform to manage applicants efficiently. It saves time and resources that would otherwise be spent on manual resume screening and communication.

For educational institutions, the system improves placement management without requiring additional manpower. The long-term benefits such as time-saving, efficiency, and better data management outweigh the initial setup cost.

Hence, the system is cost-effective and economically feasible.

2.2.3 Operational Feasibility

Operational feasibility determines whether the system will be accepted and used effectively by the users.

The Internship Portal is user-friendly and designed to be simple and easy to use for students, companies, and administrators. The interface is interactive and responsive, allowing users to access the system from laptops, tablets, and mobile devices.

Students can easily register, create profiles, search internships, and apply without

confusion. Companies can post internships and manage applications through a structured dashboard. Administrators can monitor the system and manage users efficiently.

The system reduces manual work and improves communication, which makes it highly useful in real-world scenarios. Minimal training is required for users because the interface is intuitive and similar to commonly used web applications.

The system also improves transparency by allowing users to track application status in real time. This increases user satisfaction and trust in the system.

Therefore, the system is operationally feasible and can be successfully implemented in educational institutions and organizations.

2.3 Architecture Design

For the Internship Portal, we have chosen a Three-Tier Architecture. This is a simple and reliable way to build web applications because it keeps different parts of the system separate from each other. This makes the platform easier to build, update, and fix whenever needed.

The three tiers are as follows:

1. Presentation Layer (Frontend)

This is what users actually see and use.

We built this using React.js. It shows everything on the screen – internship listings, application forms, dashboards, messages, and buttons. When a student clicks "Apply Now" or a company clicks "Post Internship," the frontend sends a request to the backend and waits for a reply.

What it does:

- Shows internship cards with all details
- Displays different dashboards for students, companies, and admins
- Handles login and registration forms
- Manages application forms where students upload resumes
- Shows messages and notifications in real time
- Works on mobile phones, tablets, and computers

2. Application Layer (Backend)

This is where all the logic and decision-making happens

We use Java Spring Boot for this layer. When the frontend sends a request (like "apply for this internship" or "accept this student"), the backend processes it, checks if everything is valid, talks to the database, and sends back the result.

What it does:

- Verifies login credentials
- Lets companies add, edit, or delete internships
- Saves student applications along with their resumes
- Processes accept/reject decisions with comments
- Handles messages between admin and users
- Makes sure students can only see student things and company can only see company things
- Serves uploaded resumes when someone wants to download them

3. Database Layer

This is where all the data lives.

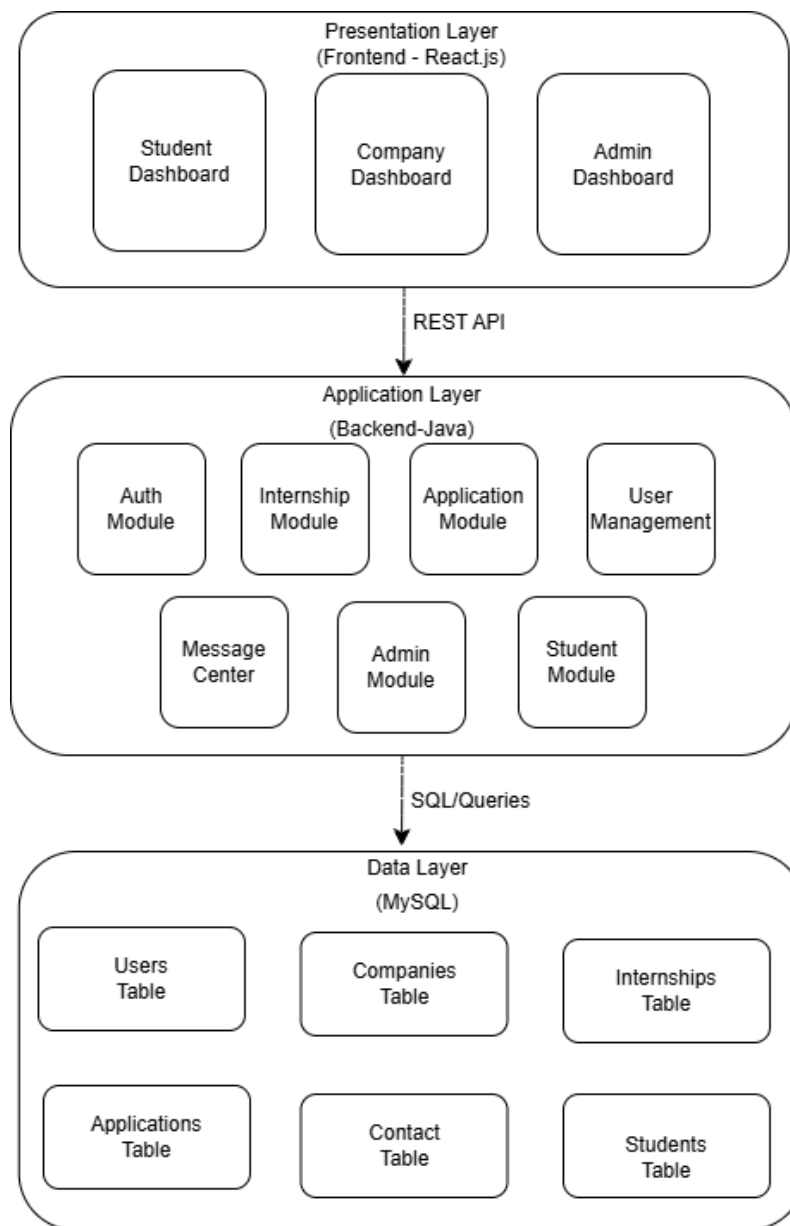
We use MySQL to store everything. It keeps user accounts, student profiles, company details, internship postings, applications, messages, and notifications safe and organized.

What it stores:

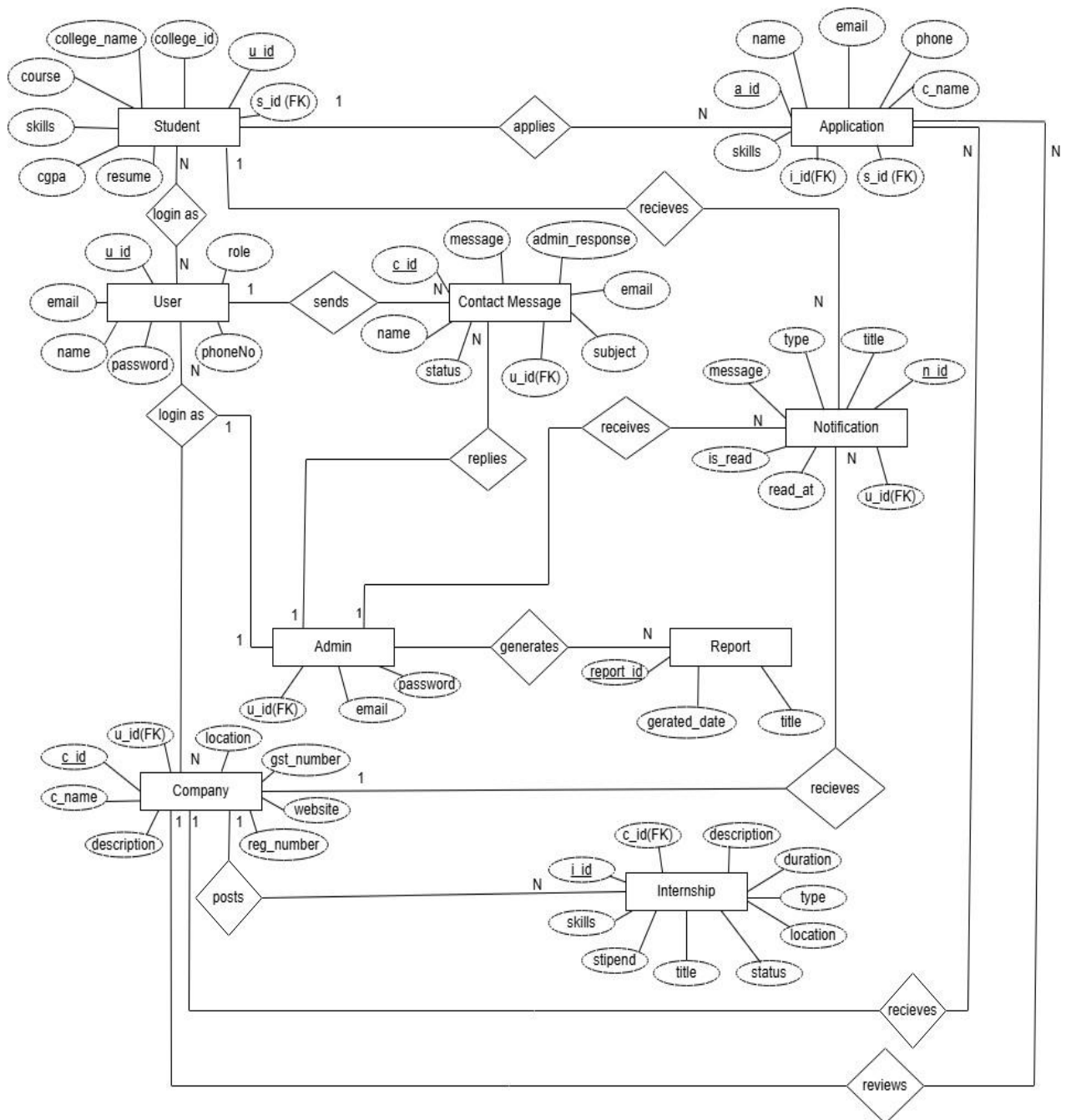
- User information (name, email, password, phone number, role)
- Student education details (college, course, branch, year, CGPA, skills)
- Company information (company name, GST number, location, website)
- Internship details (title, description, skills, stipend, duration, status)
- Application records (which student applied for which internship, status, feedback)
- Message conversations between admin and users
- Notifications for important updates

This setup works really well. If we want to change how the website looks, we only need to update the frontend. If we want to add new features or fix bugs, we can change the backend without touching the design. And if we need to organize data differently, we can update the database without affecting the other layers.

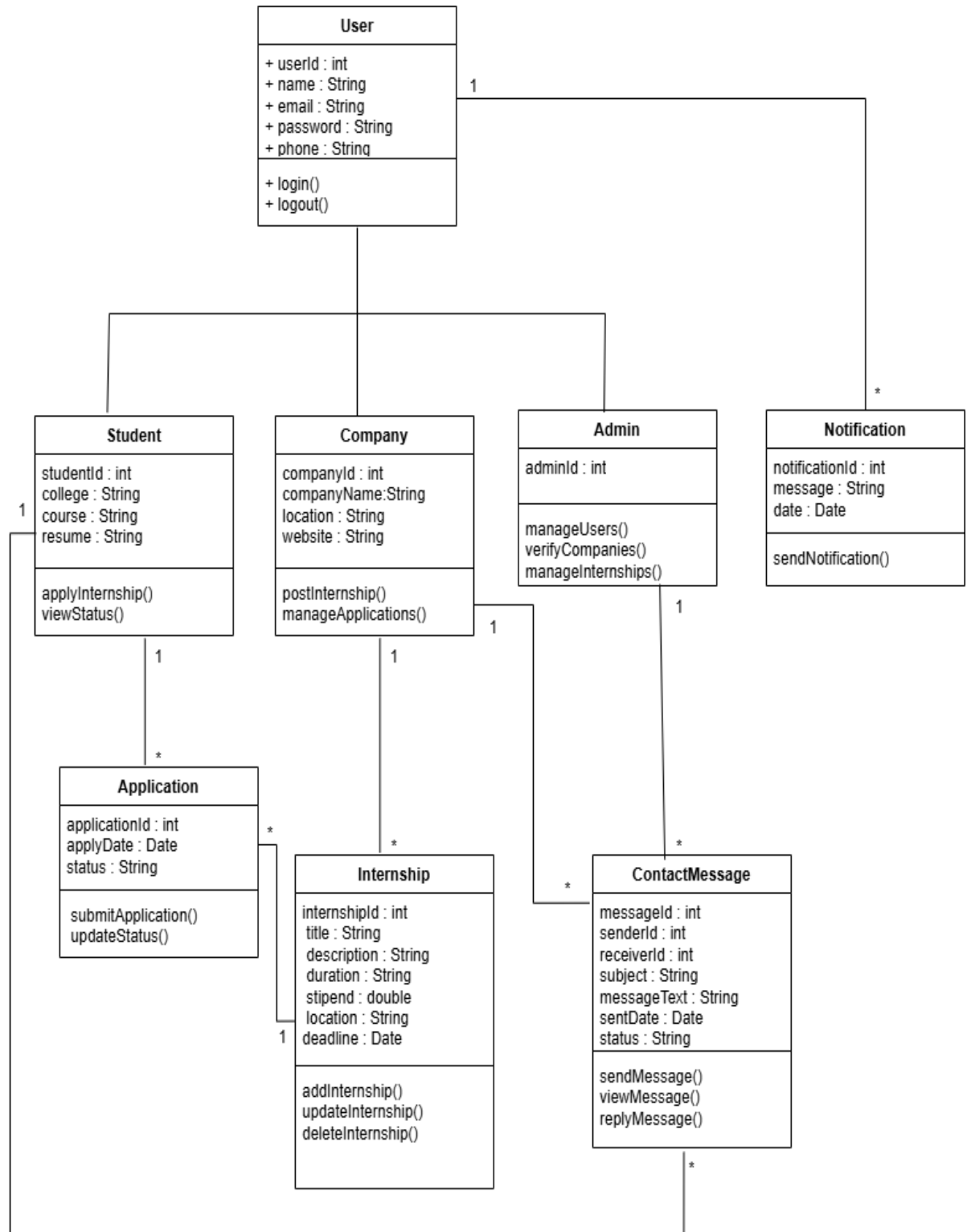
- **Architecture Diagram:**



2.4 Entity Relationship Diagram (ERD)

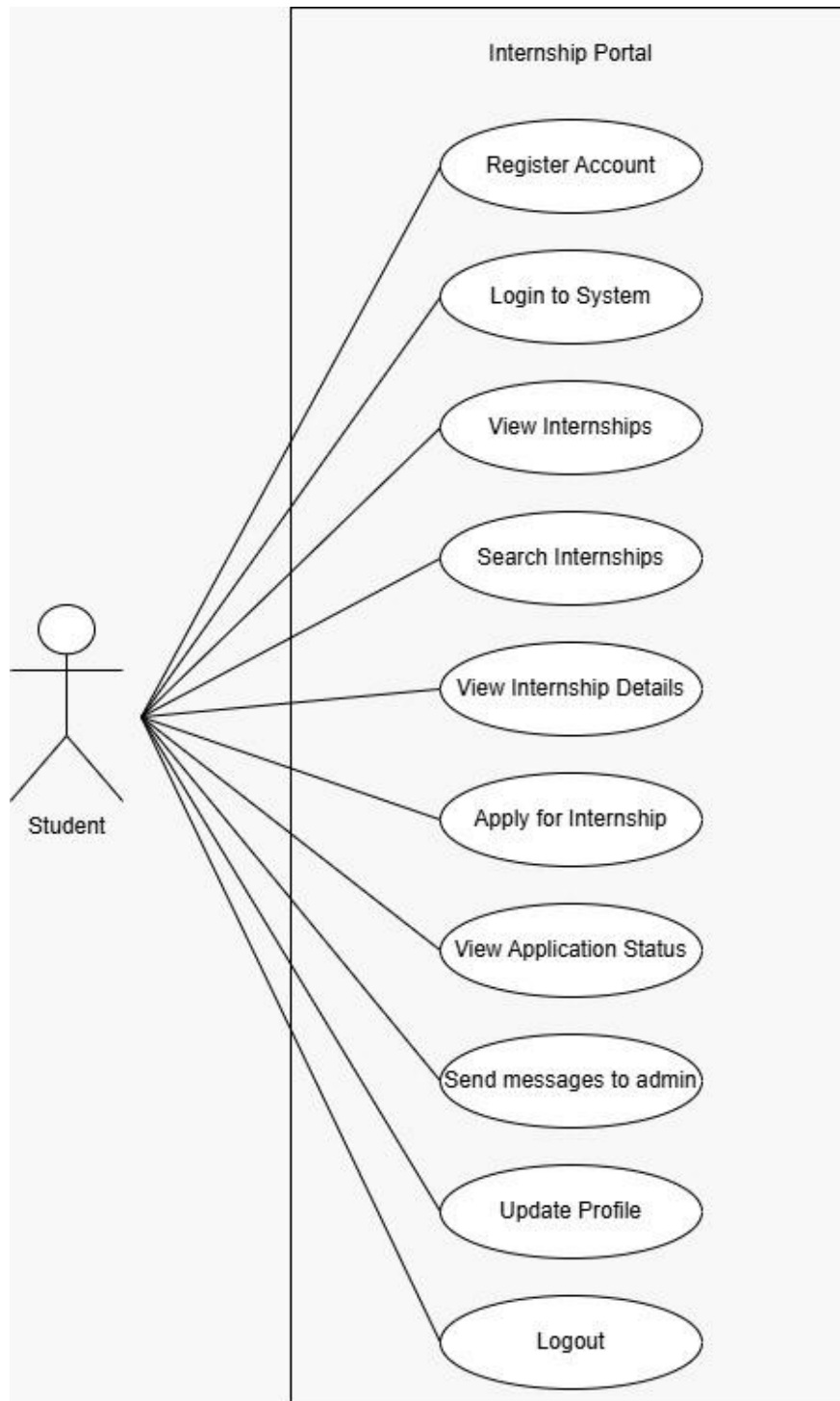


2.5 Class Diagram

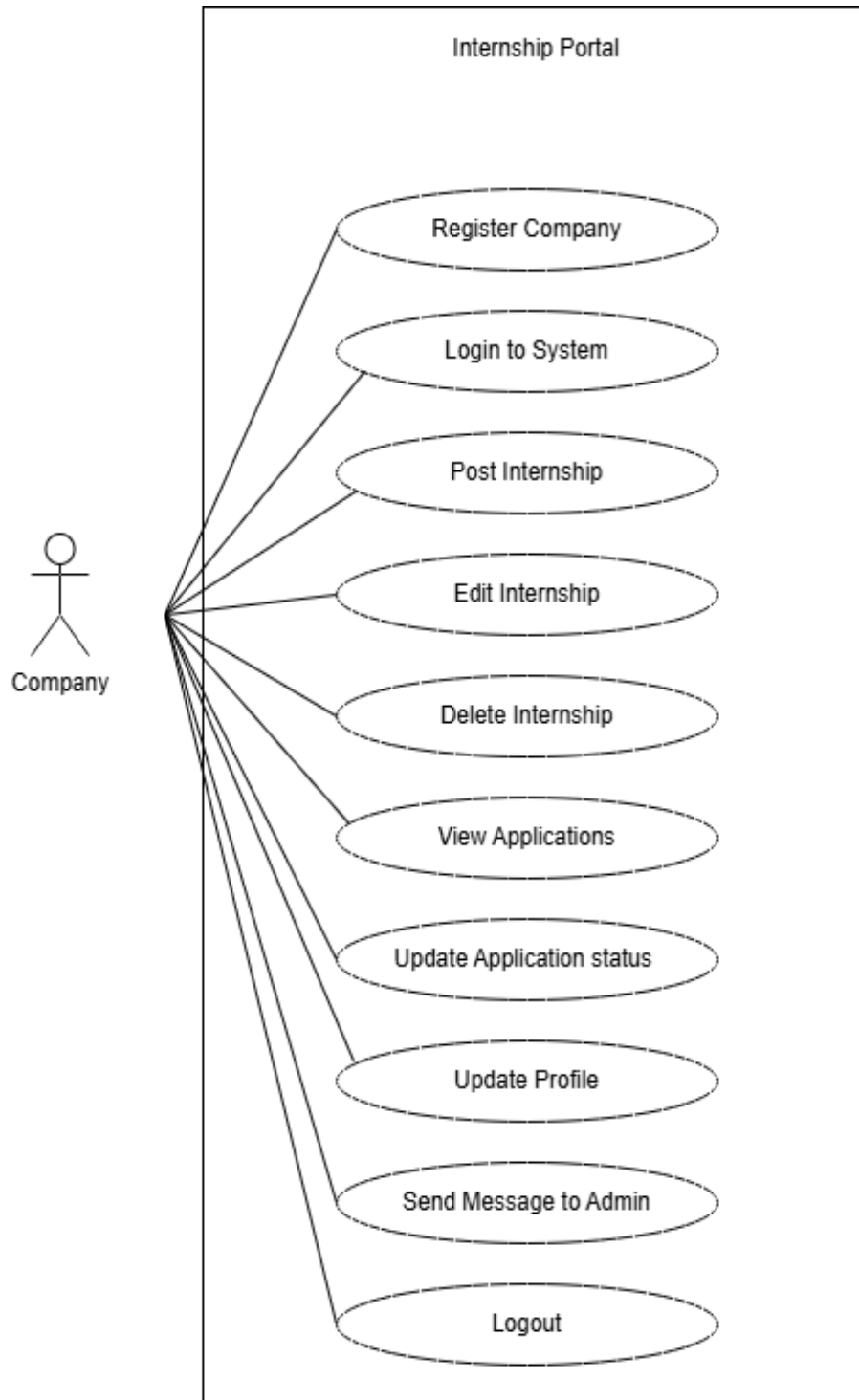


2.6 Use Case Diagrams -:

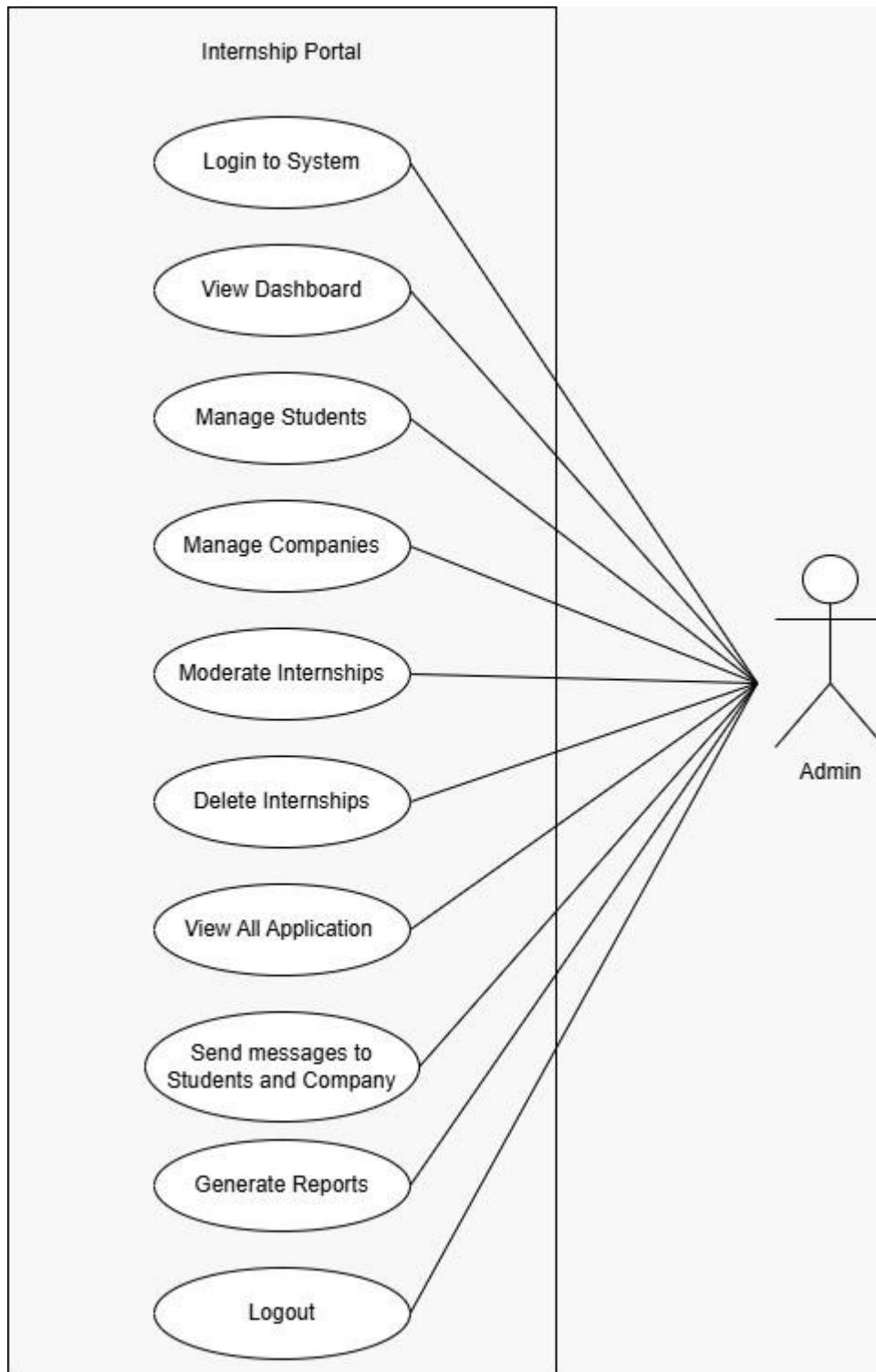
2.6.1 Student Use Case Diagram -:



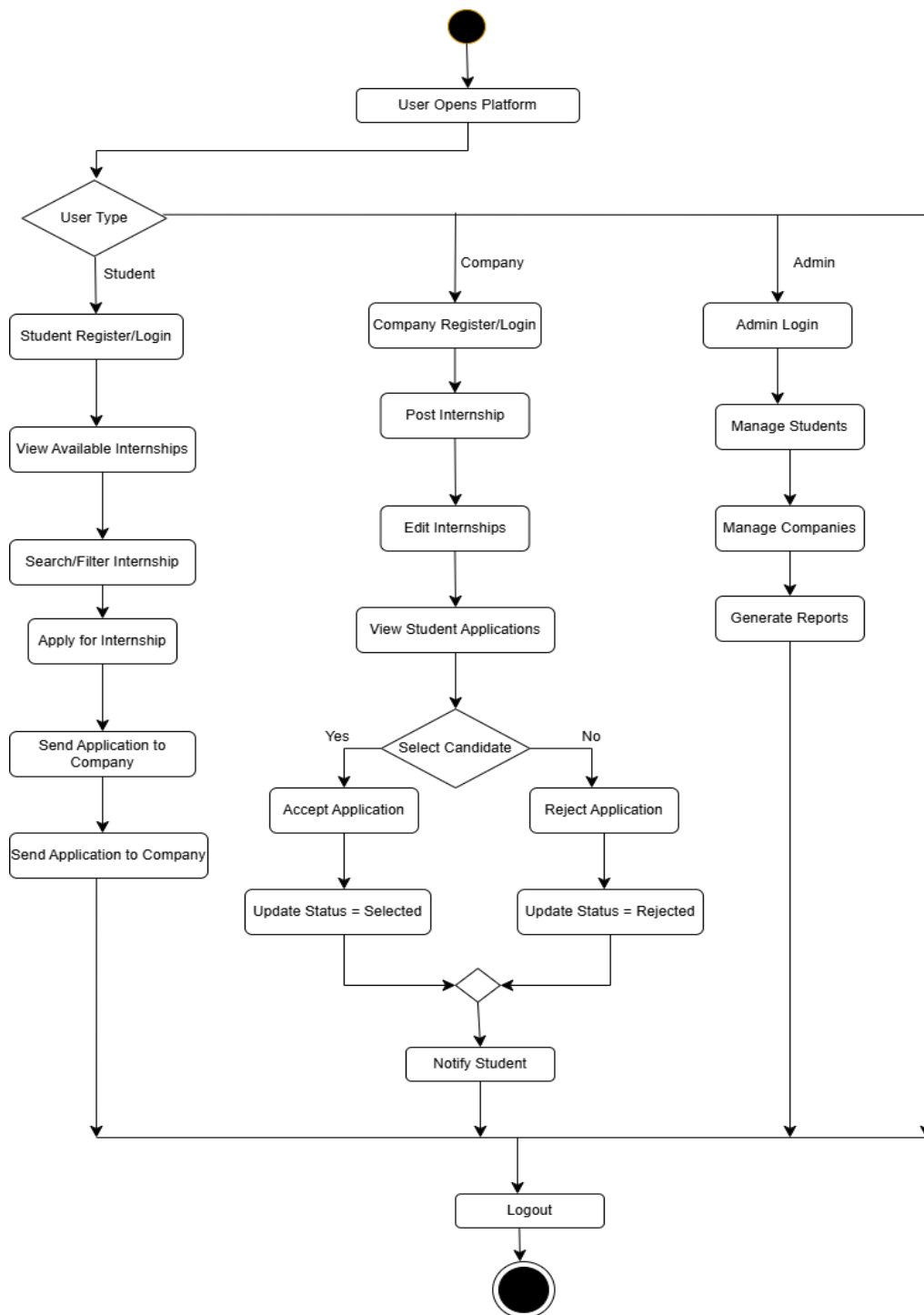
2.6.2 Company Use case Diagram :-



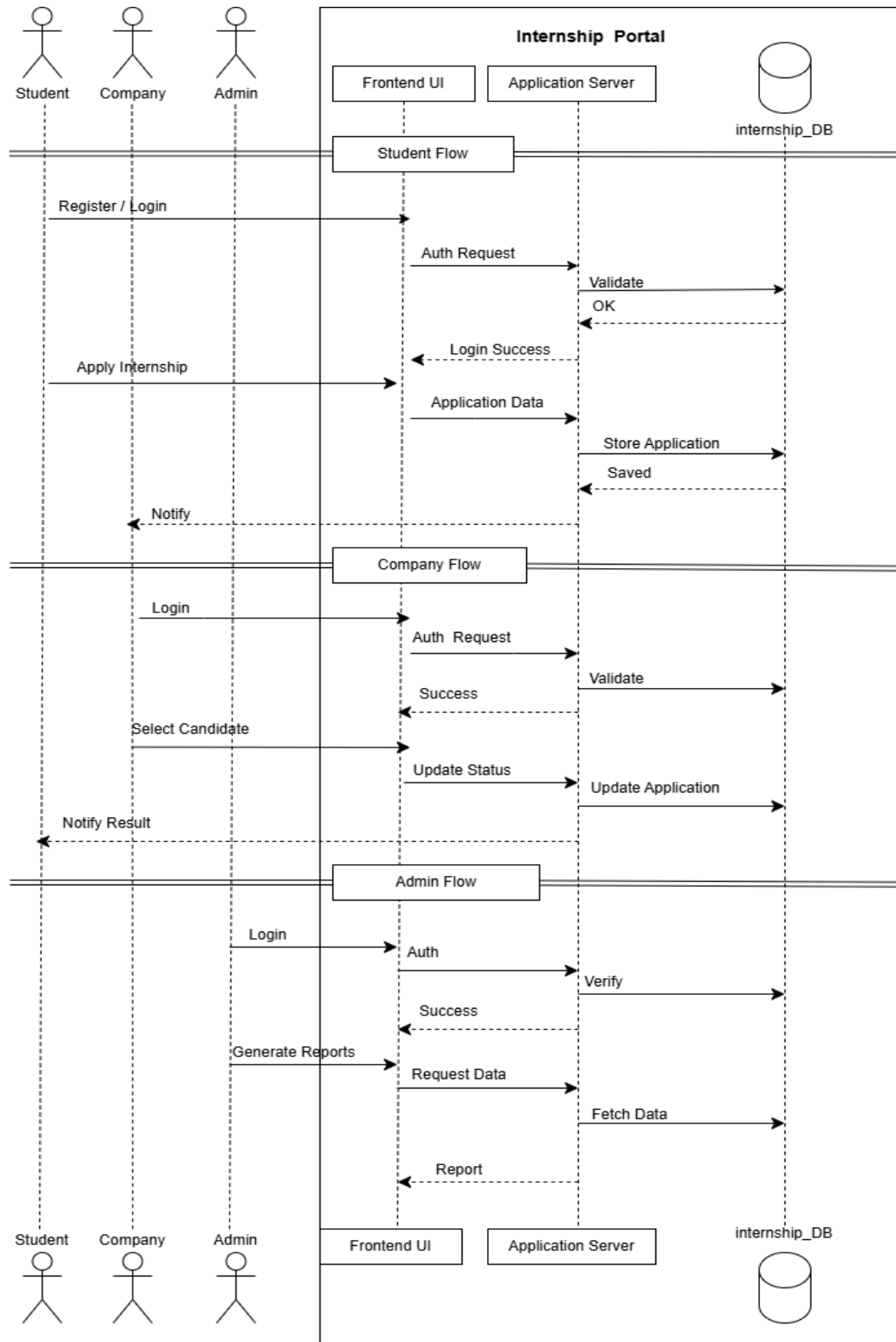
2.6.3 Admin Use case diagram :-



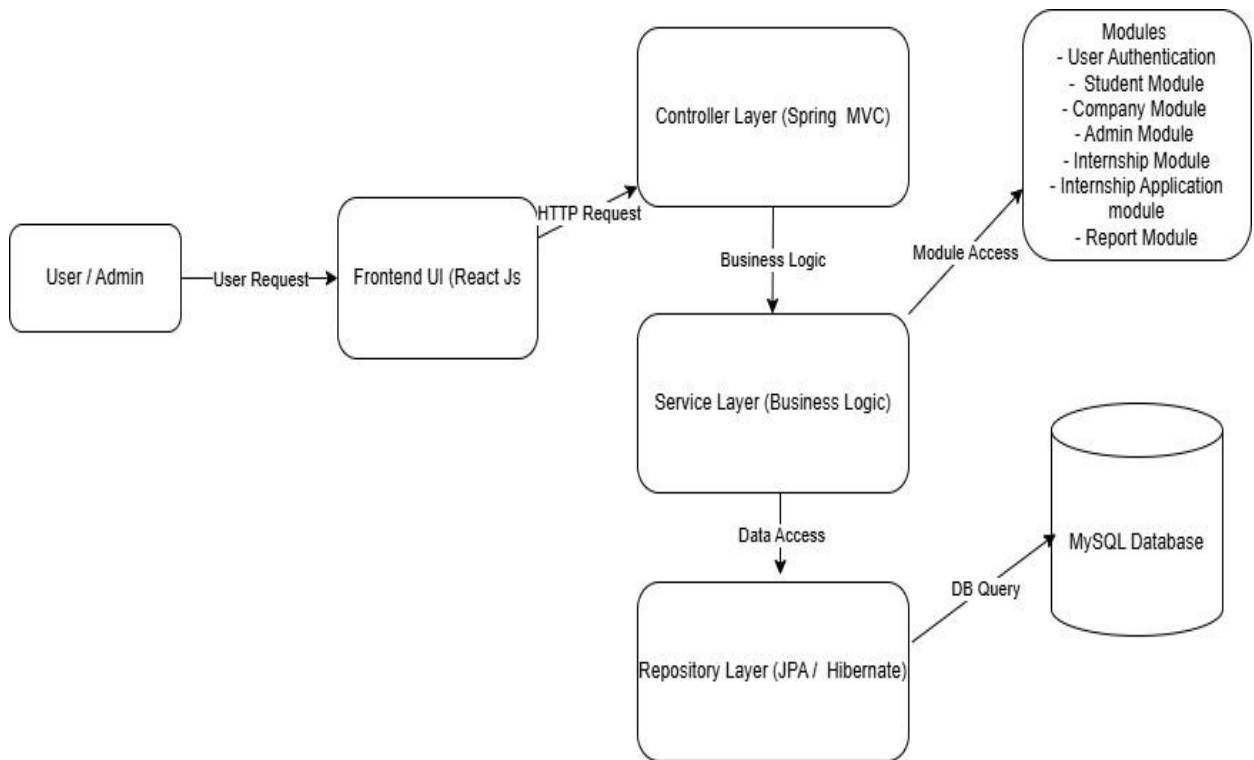
2.7 Activity Diagram -:



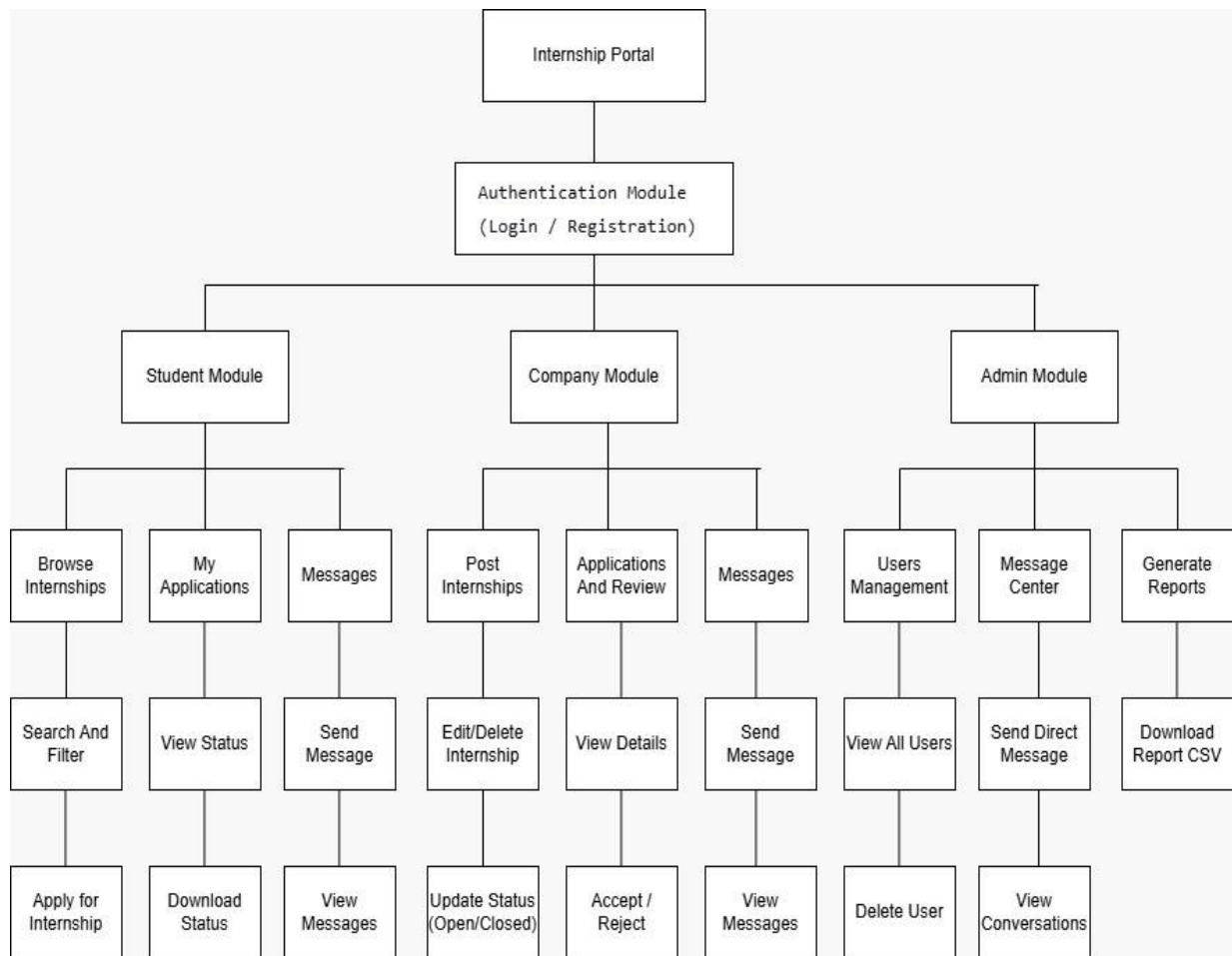
2.8 Sequence Diagram -:



2.9 Component Diagram -:



2.10 Module Hierarchy Diagram -:



2.11 Table Design

The Internship Portal web-application uses a MySQL relational database to store all system data including users, internships, and registrations.

User Table

Field Name	Data Type	Key	Description
id	bigint	PK	Unique User ID(Auto Increment)
email	varchar(255)	UNIQUE	User Email
full_name	varchar(255)	-	Full Name
password	varchar(255)	-	User Password
phone	varchar(255)	-	User Mobile Number
role	enum('ADMIN','COMPANY','STUDENT')	-	User Role
registered_at	datetime(6)	-	Registration Date & Time

Notification Table

Field Name	Data Type	Key	Description
id	bigint	PK	ID
created_at	datetime(6)	-	Created
is_read	bit(1)	-	Read Status
message	varchar(500)	-	Message
read_at	datetime(6)	-	Read At
related_id	varchar(255)	-	Related ID
title	varchar(255)	-	Title
type	varchar(255)	-	Type
user_id	bigint	FK	User

Company Table

Field Name	Data Type	Key	Description
id	bigint	PK	ID
company_name	varchar(255)	-	Company
description	text	-	Description
location	varchar(255)	-	Location
website	varchar(255)	-	Website
user_id	bigint	UNIQUE	User
logo	varchar(255)	-	Logo
gst_number	varchar(255)	-	GST
registration_number	varchar(255)	-	Registration

Student Table

Field Name	Data Type	Key	Description
id	bigint	PK	ID
branch	varchar(255)	-	Branch
cgpa	double	-	CGPA
college_name	varchar(255)	-	College
course	varchar(255)	-	Course
skills	varchar(255)	-	Skills
year_of_study	int	-	Year
user_id	bigint	FK	User
resume_filename	varchar(255)	-	Resume File
resume_path	varchar(255)	-	Resume Path
college_id_number	varchar(255)	-	College ID

Contact Table

Field Name	Data Type	Key	Description
id	bigint	PK	ID
name	varchar(255)	-	Name
email	varchar(255)	-	Email
subject	varchar(255)	-	Subject
message	varchar(2000)	-	Message
admin_response	varchar(255)	-	Response
status	varchar(255)	-	Status
created_at	datetime(6)	-	Created
responded_at	datetime(6)	-	Responded
user_id	bigint	FK	User

Application Table

Field Name	Data Type	Key	Description
id	bigint	PK	Application ID
full_name	varchar(255)	-	Name
email	varchar(255)	-	Email
phone	varchar(255)	-	Phone
college_name	varchar(255)	-	College
branch	varchar(255)	-	Branch
course	varchar(255)	-	Course
cgpa	double	-	CGPA
skills	text	-	Skills
projects	text	-	Projects
cover_letter	text	-	Cover Letter
status	varchar(255)	-	Status
applied_at	datetime(6)	-	Applied At
internship_id	bigint	FK	Internship
student_id	bigint	FK	Student

2.12 Data Dictionary – Internship Portal

Users

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Unique User ID	bigint	20	True	False	True
email	User email address	varchar	255	False	False	True
full_name	Full name of user	varchar	255	False	False	False
password	Encrypted password	varchar	255	False	False	False
phone	Mobile number	varchar	255	False	True	False
role	User role	enum	50	False	False	False
registered_at	Registration date and time	datetime(6)	-	False	False	False

Notification

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Notification ID	bigint	20	True	False	True
created_at	Notification creation date	datetime(6)	-	False	False	False
is_read	Read status	bit(1)	1	False	False	False
message	Notification message	varchar	500	False	False	False
read_at	Notification read date	datetime(6)	-	False	True	False
related_id	Related entity ID	varchar	255	False	True	False
title	Notification title	varchar	255	False	False	False
type	Notification type	varchar	255	False	False	False
user_id	Reference to user	bigint	20	False	False	False

Company Profile

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Company profile ID	bigint	20	True	False	True
company_name	Company name	varchar	255	False	False	False
description	Company description	text	-	False	True	False
location	Company location	varchar	255	False	False	False
website	Company website	varchar	255	False	True	False
user_id	Reference user ID	bigint	20	False	False	True
gst_number	GST number	varchar	255	False	True	True
registration_number	Company reginumber	varchar	255	False	True	True

Internship

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Unique internship ID	bigint	20	True	False	True
description	Internship description	varchar	255	False	False	False
duration	Internship duration	varchar	255	False	False	False
location	Internship location	varchar	255	False	False	False
posted_at	Posted date and time	datetime(6)	-	False	False	False
required_skills	Required skills	varchar	255	False	False	False
status	Internship status	enum	50	False	False	False
stipend	Internship stipend	int	10	False	True	False
title	Internship title	varchar	255	False	False	False
company_id	Reference company ID	bigint	20	False	False	False
type	Internship type	varchar	255	False	False	False
end_date	Internship end date	date	-	False	False	False
number_of_openings	Total openings	int	10	False	False	False
start_date	Internship start date	date	-	False	False	False

Contact

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Contact query ID	bigint	20	True	False	True
name	User name	varchar	255	False	False	False
email	User email	varchar	255	False	False	False
subject	Query subject	varchar	255	False	False	False
message	Query message	varchar	2000	False	False	False
admin_response	Admin response	varchar	255	False	True	False
status	Query status	varchar	255	False	False	False
created_at	Query created date	datetime(6)	-	False	False	False
responded_at	Response date	datetime(6)	-	False	True	False
user_id	Reference to user	bigint	20	False	False	False

Student

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Student ID	bigint	20	True	False	True
branch	Student branch	varchar	255	False	False	False
cgpa	Student CGPA	double	10	False	True	False
college_name	College name	varchar	255	False	False	False
course	Course name	varchar	255	False	False	False
skills	Technical skills	varchar	255	False	True	False
year_of_study	Current study year	int	10	False	False	False
user_id	Reference user ID	bigint	20	False	False	False
resume_filename	Resume file name	varchar	255	False	True	False
resume_path	Resume storage path	varchar	255	False	True	False
college_id_number	College ID number	varchar	255	False	True	True

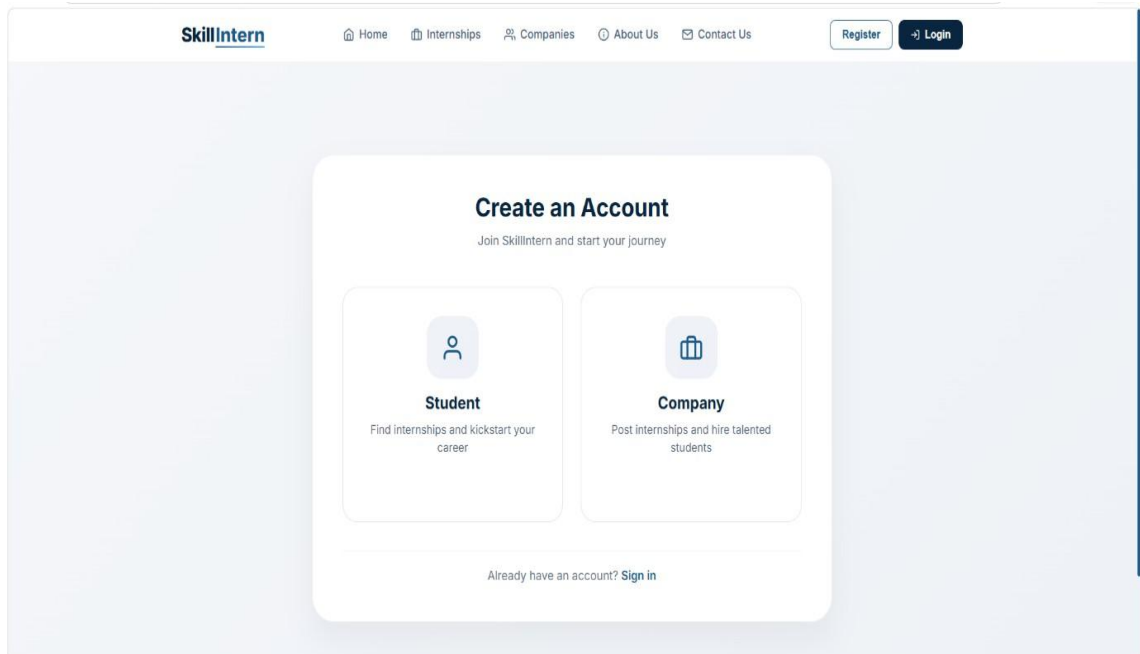
Application

Column Name	Column Description	Data Type	Length	Primary Key	Nullable	Unique
id	Application ID	bigint	20	True	False	True
full_name	Applicant name	varchar	255	False	False	False
email	Applicant email	varchar	255	False	False	False
phone	Applicant phone number	varchar	255	False	False	False
college_name	College name	varchar	255	False	False	False
branch	Branch name	varchar	255	False	False	False
course	Course name	varchar	255	False	False	False
cgpa	Applicant CGPA	double	10	False	True	False
skills	Skills details	text	-	False	True	False
projects	Project details	text	-	False	True	False
cover_letter	Cover letter	text	-	False	True	False
status	Application status	varchar	255	False	False	False
applied_at	Application date	datetime(6)	-	False	False	False
internship_id	Reference internship ID	bigint	20	False	False	False
student_id	Reference student ID	bigint	20	False	False	False

Chapter 3: Implementation

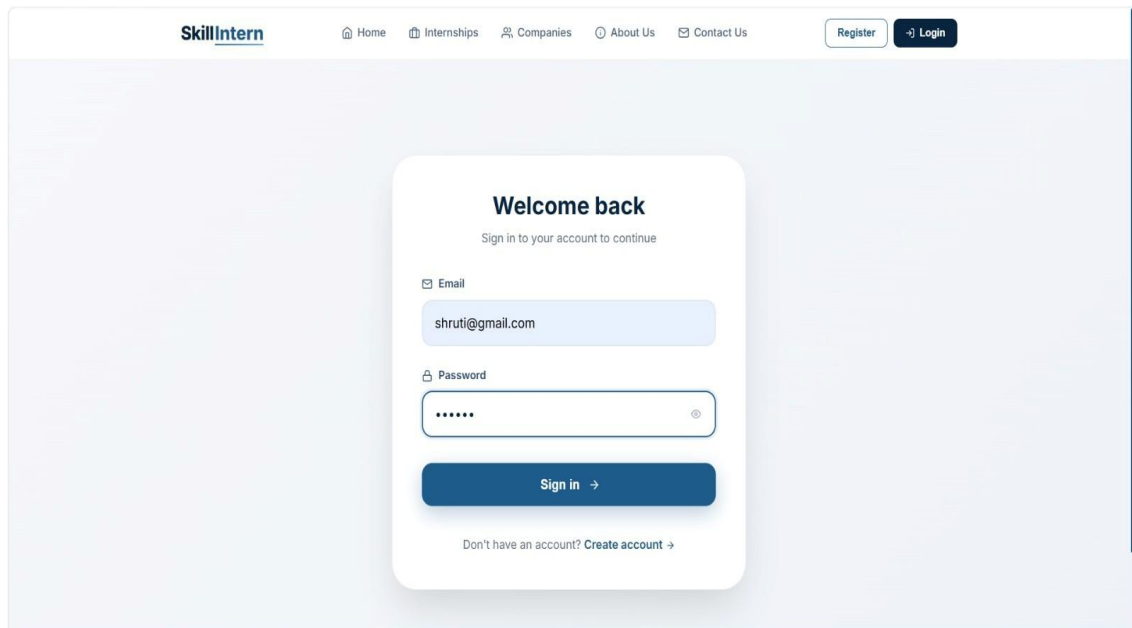
3.1 Input Screens:

- **Registration Page :**



The registration page for SkillIntern features a clean, modern design. At the top, a navigation bar includes the SkillIntern logo, links for Home, Internships, Companies, About Us, and Contact Us, and buttons for Register and Login. The main content area is centered and titled "Create an Account" with the subtitle "Join SkillIntern and start your journey". Below this, there are two distinct cards: "Student" with a person icon and the text "Find internships and kickstart your career", and "Company" with a briefcase icon and the text "Post internships and hire talented students". At the bottom of the central area, a link says "Already have an account? Sign in".

- **Login Page :**



The login page for SkillIntern maintains the same header as the registration page. The main content area is centered and titled "Welcome back" with the subtitle "Sign in to your account to continue". It features two input fields: "Email" with the placeholder "shruti@gmail.com" and "Password" with masked characters "*****". Below these fields is a prominent blue "Sign in" button with a right-pointing arrow. At the bottom, a link says "Don't have an account? Create account →".

- **Student Registration Form :**

The screenshot shows the 'Student Registration' form on the SkillIntern website. The header includes the SkillIntern logo and navigation links: Home, Internships, Companies, About Us, and Contact Us. There are 'Register' and 'Login' buttons in the top right. A 'Back to Role Selection' link is at the top left of the form area. The form title is 'Student Registration'. Below it is a section titled 'Personal Information' with a person icon. The form contains the following fields: 'FULL NAME *' with the value 'Shruti Pawar'; 'EMAIL *' with the value 'shruti@gmail.com'; 'PHONE NUMBER *' with the value '9586231457' and a note 'Enter a valid 10-digit mobile number starting with 6,7,8, or 9'; 'PASSWORD *' with the value 'Shruti@123' and a note 'Must contain uppercase, lowercase, number and special character'; and 'CONFIRM PASSWORD *' with the value 'Shruti@123'.

- **Company Registration Form :**

The screenshot shows the 'Company Registration' form on the SkillIntern website. The header is identical to the previous form. The form title is 'Company Registration'. Below it is a section titled 'Contact Person Details' with a person icon. The form contains the following fields: 'CONTACT PERSON NAME *' with the value 'Cognizant HR'; 'EMAIL *' with the value 'cognizant@gmail.com'; 'PHONE NUMBER *' with the value '9886766555' and a note 'Enter a valid 10-digit mobile number starting with 6,7,8, or 9'; 'PASSWORD *' with the value 'Cognizant@123' and a note 'Must contain uppercase, lowercase, number and special character'; and 'CONFIRM PASSWORD *' with masked characters '*****'.

- **Post New Internship Form :**

The screenshot shows a modal form titled "Post New Internship" overlaid on a blurred dashboard background. The form contains the following fields:

- INTERNSHIP TITLE ***: A text input field containing "Frontend Developer".
- DESCRIPTION ***: A text area containing "Frontend Developer with basic knowledge of HTML, CSS, JavaScript and MySQL".
- REQUIRED SKILLS ***: A text input field containing "HTML, CSS, JavaScript, MySQL".

- **Contact Form :**

The screenshot shows the "Send us a Message" contact form on the SkillIntern website. The form includes the following elements:

- Header**: "Send us a Message" with a subtext "Fill out the form below and we'll get back to you shortly".
- Form Fields**:
 - FULL NAME ***: Input field with "Shruti pawar".
 - EMAIL ADDRESS ***: Input field with "shruti@gmail.com".
 - SUBJECT ***: Input field with "Unable to upload resume".
 - MESSAGE ***: Text area with "I am unable to upload resume in my profile".
- Character Count**: "42/2000 characters" below the message field.
- Submit Button**: "Send Message" button at the bottom.
- Business Hours**: A section on the right showing:
 - Monday - Friday: 9:00 AM - 6:00 PM
 - Saturday: 10:00 AM - 4:00 PM
 - Sunday: Closed
- Quick Answers**: A section on the right with FAQs:
 - How do I apply for internships?**: Create a student account, browse internships, and click "Apply Now".
 - How do I post internships?**: Register as a company, complete your profile, and post internships.
 - Is SkillIntern free?**: Yes! Both students and companies can use SkillIntern for free.

- **Application Form :**

Apply for Full Stack Web Developer Intern
• at Infosys

1 Personal & Academic 2 Resume & Questions

Personal Information

👤 FULL NAME * Pooja Kale

✉ EMAIL * pooja@gmail.com

📞 PHONE * 8668473513

Apply for Full Stack Web Developer Intern
• at Infosys

1 Personal & Academic 2 Resume & Questions

Resume Upload *

📎 Choose Resume (PDF/DOC)

Max size: 5MB, Formats: PDF, DOC, DOCX

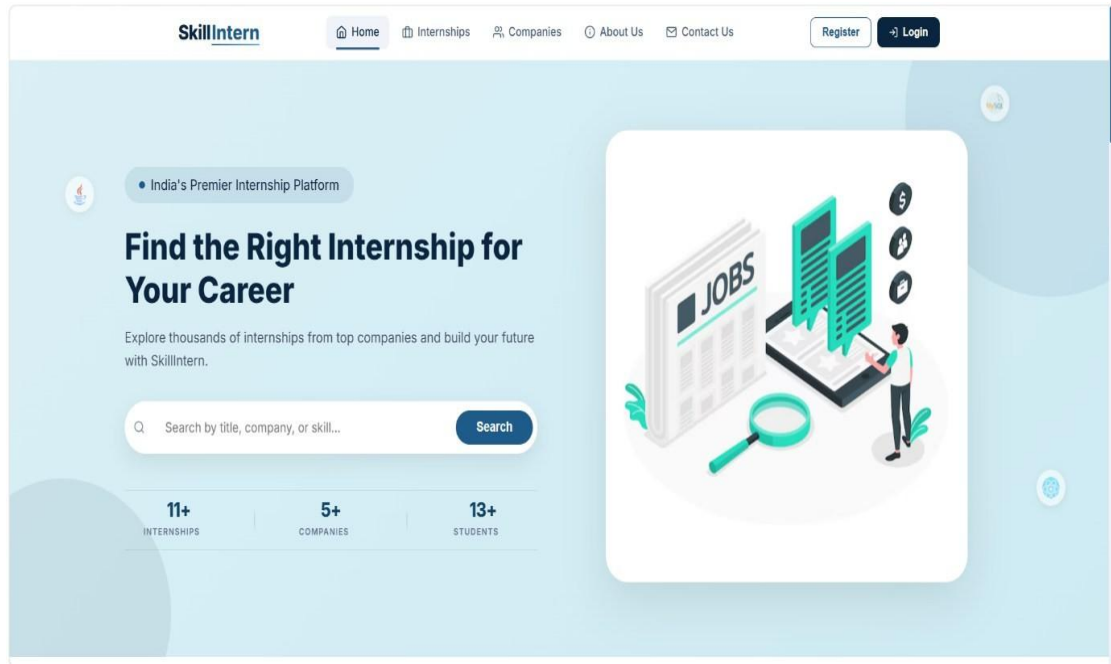
Application Questions

WHY SHOULD WE HIRE YOU? *

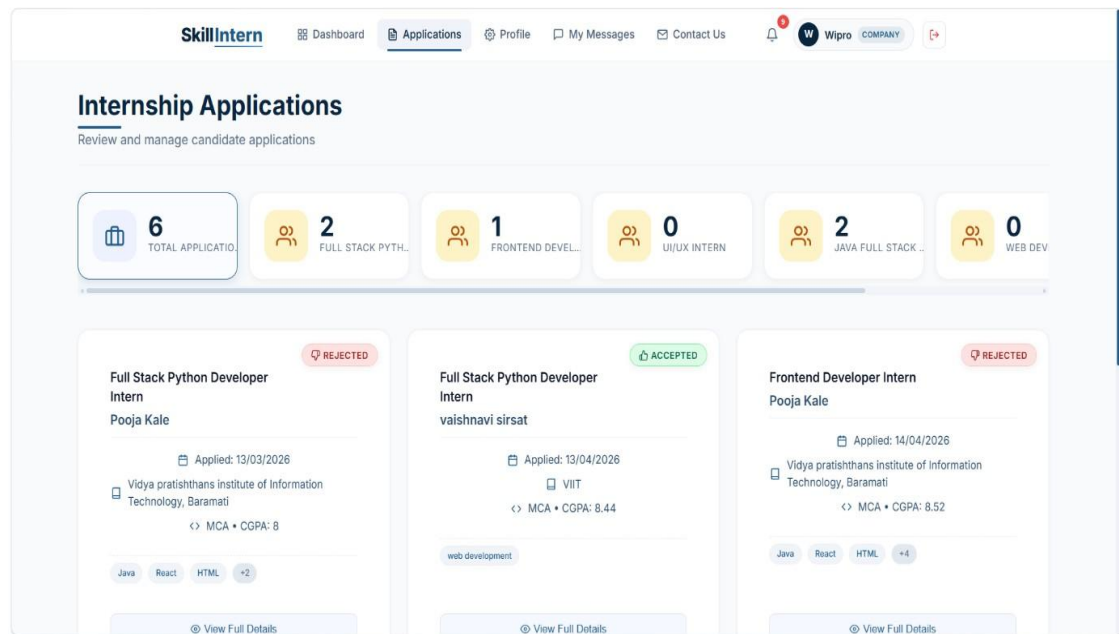
Explain why you're the best candidate for this internship

3.2 Output Screens / Reports :-

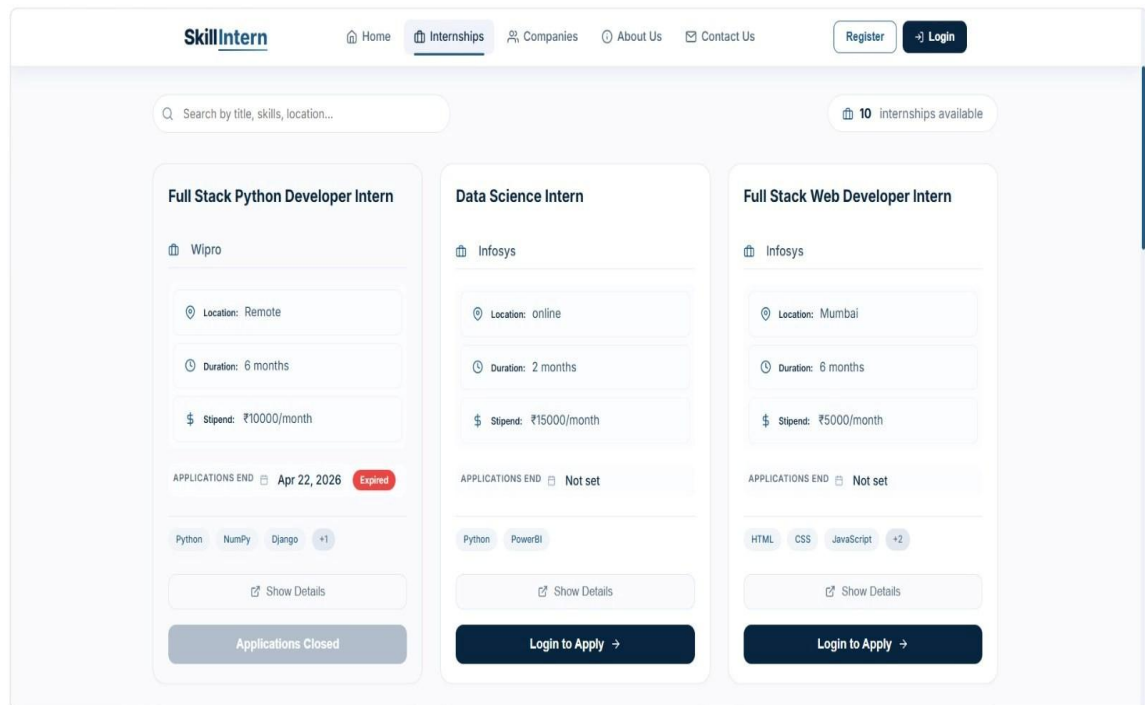
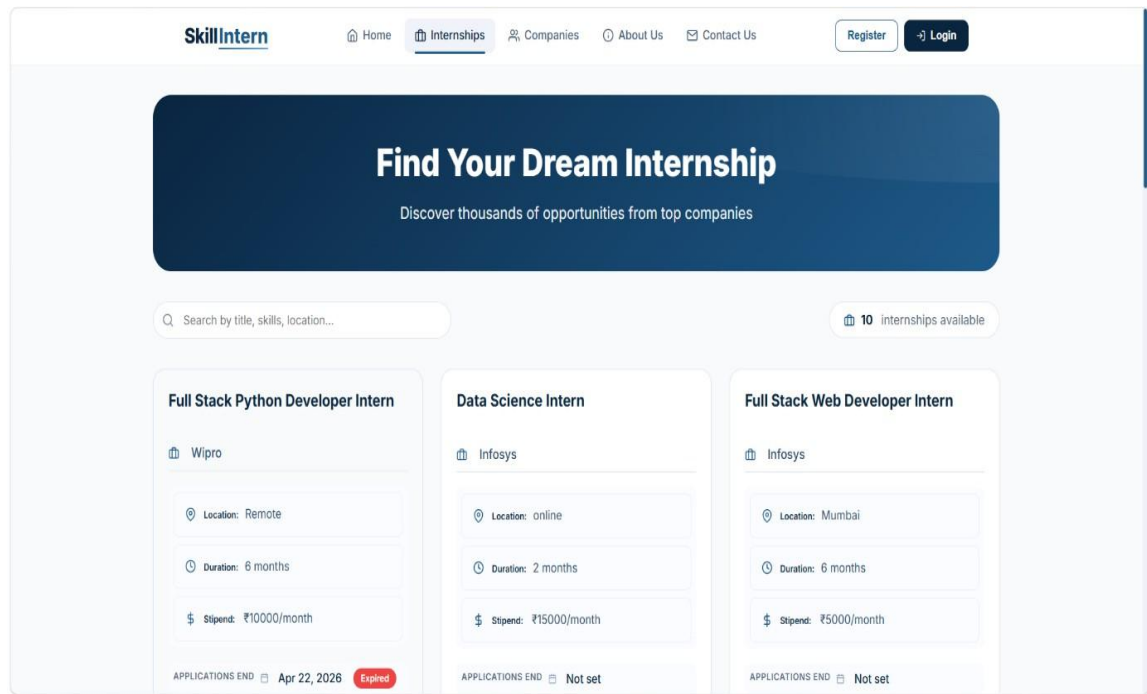
- **Home Page :**



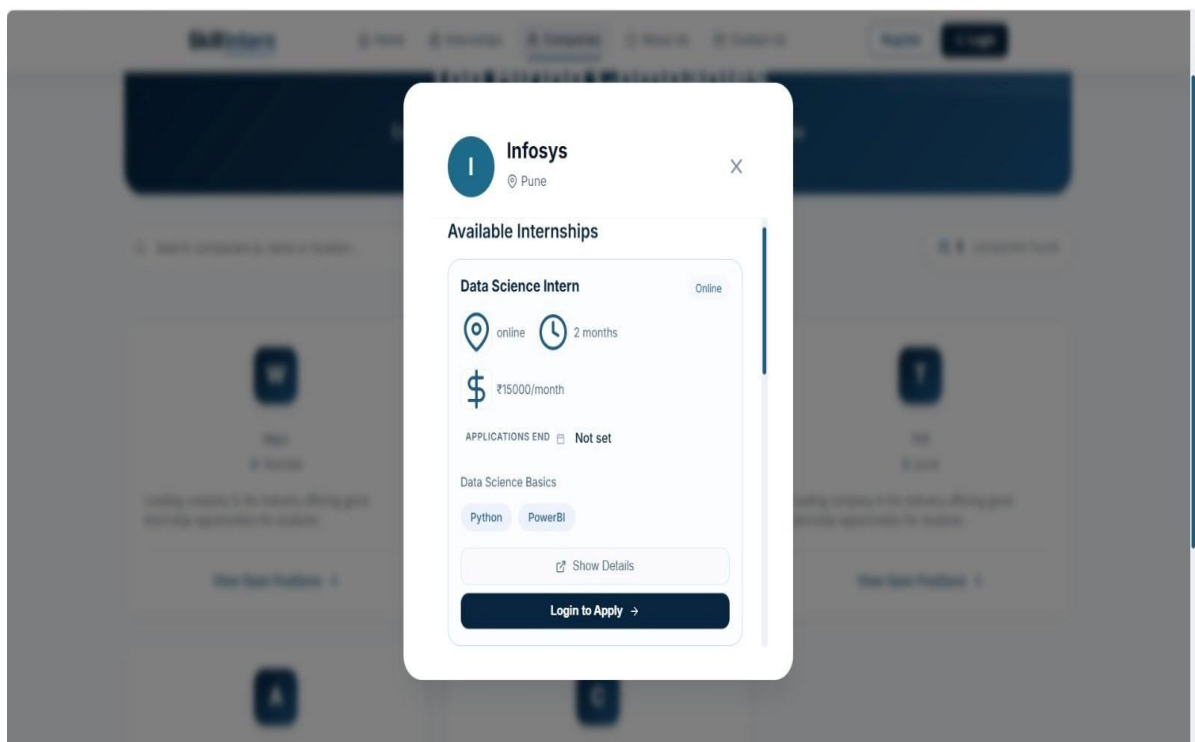
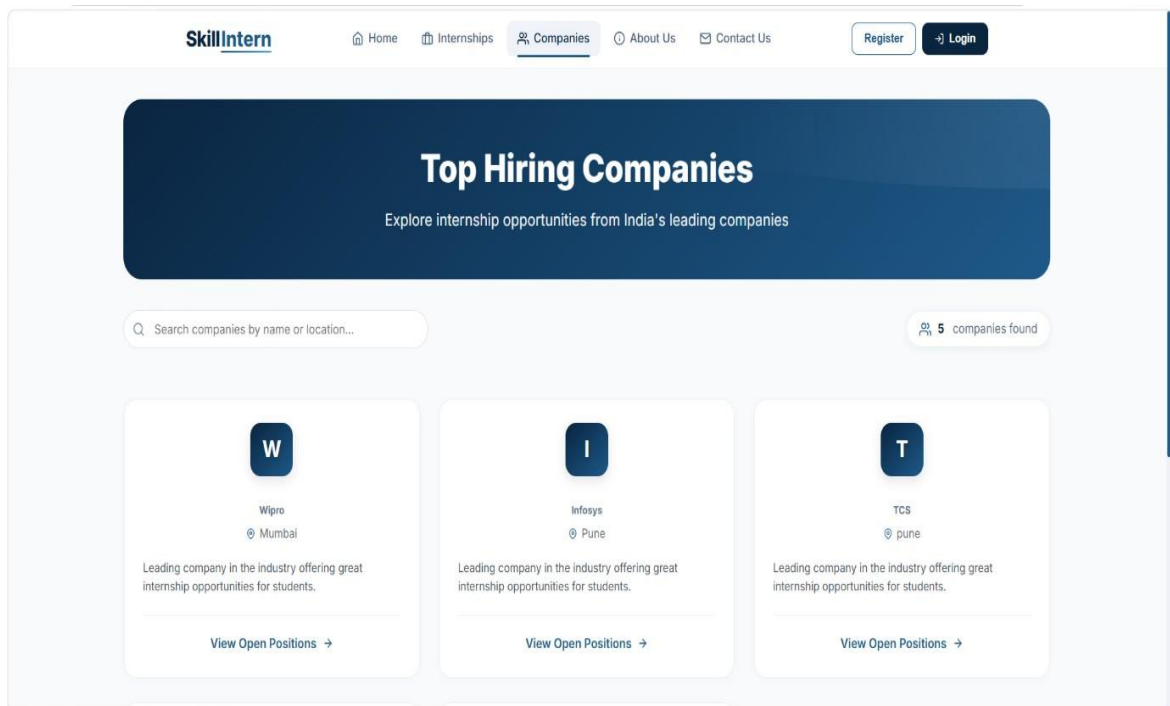
- **Application Status:**



- **Internships Page :**



- **Companies Page :**



- **Reports:**

Admin Dashboard

System Administrator
admin@skillintern.com

Admin Panel

Dashboard

Users

Students

Companies

Internships

Applications

Contact Messages

Reports

Reports

Generate and download reports

REPORT TYPE

DATE RANGE

Generate

Download

Internships Report

2026-04-13 to 2026-05-13

11
TOTAL INTERNSHIPS

80,500
TOTAL STIPEND VALUE

76
TOTAL OPENINGS

10
OPEN INTERNSHIPS

7,318.182
AVG STIPEND

1
CLOSED INTERNSHIPS

posted At	end Date	company Name	stipend	location	id	title	applications Count	status
2026-03-06	2026-06-25	Wipro	10000	Remote	3	Full Stack Python Developer Intern	2	OPEN
2026-03-06	Not specified	Infosys	15000	online	4	Data Science Intern	5	OPEN

Admin Dashboard

System Administrator
admin@skillintern.com

Admin Panel

Dashboard

Users

Students

Companies

Internships

Applications

Contact Messages

Reports

Reports

Generate and download reports

REPORT TYPE

DATE RANGE

Generate

Download

Users Report

2026-04-13 to 2026-05-13

19
TOTAL USERS

1
TOTAL ADMINS

0
NEW USERS THIS MONTH

13
TOTAL STUDENTS

5
TOTAL COMPANIES

role	registered At	full Name	id	email
STUDENT	Not specified	Pooja Kale	1	pooja@gmail.com
STUDENT	Not specified	Shruti pawar	2	shruti@gmail.com
COMPANY	Not specified	Wipro HR	3	wipro@gmail.com
COMPANY	Not specified	Infosys	4	infosys@gmail.com

Admin Dashboard

System Administrator
admin@skillintern.com

Admin Panel

Dashboard

Users

Students

Companies

Internships

Applications

Contact Messages

Reports

Reports

Generate and download reports

REPORT TYPE

DATE RANGE

Students

Last 30 Days

Generate

Download

Students Report

2026-04-13 to 2026-05-13

46.154

PLACEMENT RATE

8.355

AVG CGPA

13

TOTAL STUDENTS

0.923

AVG APPLICATIONS PER STUDENT

12

TOTAL APPLICATIONS

college	accepted Count	name	course	id	cgpa	applications Count	email
Vidya pratishthans institute of Information Technology, Baramati	1	Pooja Kale	MCA	1	8.56	4	pooja@gmail.com
Vidya pratishthans institute of Information Technology, Baramati	1	Shruti pawar	MCA	2	8.5	2	shruti@gmail.com
Vidya pratishthans institute of Information Technology, Baramati	0	Sita Singh	MCA	4	8.2	0	sita@gmail.com
Vidya pratishthans institute of Information Technology, Baramati	0	Ram Bhosale	MCA	5	8.24	0	ram@gmail.com

Admin Dashboard

System Administrator
admin@skillintern.com

Admin Panel

Dashboard

Users

Students

Companies

Internships

Applications

Contact Messages

Reports

Reports

Generate and download reports

REPORT TYPE

DATE RANGE

Applications

Last 30 Days

Generate

Download

Applications Report

2026-04-14 to 2026-05-14

4

PENDING APPLICATIONS

6

ACCEPTED APPLICATIONS

2

REJECTED APPLICATIONS

0

SHORTLISTED APPLICATIONS

12

TOTAL APPLICATIONS

50

ACCEPTANCE RATE

student Email	Internship Title	company Name	student Name	id	cgpa	applied At	status
pooja@gmail.com	Full Stack Python Developer Intern	Wipro	Pooja Kale	5	8	2026-03-13	REJECTED
pooja@gmail.com	Data Science Intern	Infosys	Pooja Kale	6	8.52	2026-03-15	PENDING

3.3 Sample Code:-

Home Page:

```
import React, { useState, useEffect } from 'react';
import { Link, useNavigate } from 'react-router-dom';
import { FiSearch, FiArrowRight, FiMapPin, FiClock, FiDollarSign, FiBriefcase, FiUsers,
FiAward } from 'react-icons/fi';
import homeService from '../services/homeService';
import './HomePage.css';
const HomePage = () => {
  const navigate = useNavigate();
  const [companies, setCompanies] = useState([]);
  const [internships, setInternships] = useState([]);
  const [searchTerm, setSearchTerm] = useState("");
  const [stats, setStats] = useState({
    internships: 0,
    companies: 0,
    activeStudents: 0
  });
  setCompanies(companiesData || []);
  setInternships(internshipsData || []);
  setStats({
    internships: statsData?.internships || internshipsData?.length || 0,
    companies: statsData?.companies || companiesData?.length || 0,
    activeStudents: statsData?.students || statsData?.activeStudents || 12500
  });
} catch (error) {
  console.error('Error fetching home data:', error);
  setStats({
    internships: 0,
    companies: 0,
    activeStudents: 12500
  });
}
```

```

});
} finally {
  setLoading(false);
}
};

<section className="hero-section">
  <div className="hero-container">
    <div className="hero-left">
      <div className="hero-badge">
        <span className="badge-dot"></span>
        India's Premier Internship Platform
      </div>
      <h1> Find the Right Internship for Your Career
    </h1>
    <form onSubmit={handleSearch} className="hero-search">
      <div className="search-wrapper">
        <FiSearch className="search-icon" />
        <input
          type="text"
          placeholder="Search by title, company, or skill..."
          value={searchTerm}
          onChange={(e) => setSearchTerm(e.target.value)}
        />
        <button type="submit">Search</button>
      </div>
    </form>
  </div>
</div>
</section>

<section className="featured-section">
  <div className="container">

```

```

    <div className="section-header">
    </div>
    <div className="internships-grid">
      {internships.slice(0, 6).map((internship) => (
        <div key={internship.id} className="internship-card">
          <h3>{internship.title}</h3>
          <p className="company-name">{internship.companyName}</p>
          <div className="card-details">
            <span><FiMapPin /> {internship.location || 'Remote'}</span>
            <span><FiClock /> {internship.duration || '3 months'}</span>
            <span><FiDollarSign /> ₹ {internship.stipend}/mo</span>
          </div>
          <div className="card-skills">
            {internship.requiredSkills?.split(',').slice(0, 3).map((skill, i) => (
              <span key={i}>{skill.trim()}</span>
            ))}
          </div>
          <button className="apply-btn" onClick={() => navigate('/register')}>
            Apply Now <FiArrowRight />
          </button>
        </div>
      ))}
    </div>
  </div>
</section>

);
};
export default HomePage;

```

AuthController.java:

```
package com.example.demo.controller;

import com.example.demo.dto.LoginRequest;
import com.example.demo.dto.LoginResponse;
import com.example.demo.dto.RegisterRequest;
import com.example.demo.service.AuthService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.*;
import java.util.HashMap;
import java.util.Map;

@CrossOrigin(origins = {"http://localhost:3000", "http://localhost:5173"})
@RestController
@RequestMapping("/api/auth")
public class AuthController {

    @Autowired
    private AuthService authService;

    @PostMapping("/login")
    public ResponseEntity<?> login(@RequestBody LoginRequest request) {
        try {
            LoginResponse response = authService.login(request.getEmail(),
request.getPassword());

            System.out.println("=== Login Response being sent to frontend ===");
            System.out.println("userId: " + response.getUserId());
            System.out.println("email: " + response.getEmail());
            System.out.println("fullName: " + response.getFullName());
            System.out.println("role: " + response.getRole());
            System.out.println("phone: " + response.getPhone());

            System.out.println("=====");
            return ResponseEntity.ok(response);
        }
    }
}
```

```

    } catch (Exception e) {
        Map<String, String> error = new HashMap<>();
        error.put("error", e.getMessage());
        return ResponseEntity.badRequest().body(error);
    }
}

@PostMapping("/register")
public ResponseEntity<?> register(@RequestBody RegisterRequest request) {
    try {
        LoginResponse response = authService.register(request);
        System.out.println("=== Registration Response ===");
        System.out.println("userId: " + response.getUserId());
        System.out.println("email: " + response.getEmail());
        System.out.println("fullName: " + response.getFullName());
        System.out.println("role: " + response.getRole());
        System.out.println("phone: " + response.getPhone());
        System.out.println("=====");
        return ResponseEntity.ok(response);
    } catch (Exception e) {
        Map<String, String> error = new HashMap<>();
        error.put("error", e.getMessage());
        return ResponseEntity.badRequest().body(error);
    }
}
}

```


Chapter 4: Testing

4.1 Test Strategy

Testing is a vital part of our development process, designed to ensure that the SkillIntern Internship Portal is reliable, secure, and performs exactly as expected by students, companies, and administrators. Our strategy focuses on identifying and fixing issues before the platform reaches our users. We follow a multi-layered approach: starting with Unit Testing to verify small blocks of code, moving to Integration Testing to ensure that different modules like the Frontend (React) and Backend (Spring Boot) communicate without data loss, and finally System Testing where the entire platform is evaluated as a complete product.

We aim to validate all core functionalities, including:

- User Authentication: Registration, login, role-based access for Students, Companies, and Admins
- Internship Management: Posting, editing, and closing internships by companies
- Application Process: Students browsing and applying to internships with resume upload
- Review System: Companies reviewing, accepting, or rejecting applications with feedback
- Messaging System: Direct communication between Admin and users
- Profile Management: Updating personal and company information

By using a combination of automated backend tests and manual frontend testing, we guarantee a professional, bug-free experience for every user.

4.2 Unit Test Plan

Our unit testing plan focuses on verifying that the smallest individual parts of our application work correctly in isolation. By testing components separately, we can catch logic errors early in the development lifecycle, which makes maintenance much easier.

Frontend Components:

- Testing individual UI elements like login forms, registration forms, internship cards,

- application modals, and search bars in React to ensure they capture and send user data accurately
- Testing navigation and routing to ensure users are redirected to correct dashboards based on their role (Student, Company, Admin)
 - Testing form validations (email format, password strength, phone number validation) before data is sent to the backend

Backend API Modules:

- Every API endpoint in our Spring Boot application is tested to ensure it correctly handles inputs, performs database operations, and returns valid responses (e.g., HTTP 200 OK for success, 400 for bad requests, 401 for unauthorized access)
- Testing service layer methods like login(), register(), applyForInternship(), reviewApplication(), and sendUserReply() for correct business logic
- Testing repository layer for database queries and relationships

Tools Used:

- VS Code Debugger – To trace logic errors during code execution
- Postman – To simulate real HTTP requests to our backend and verify API responses before the frontend is fully connected
- Browser DevTools – To inspect network requests, console logs, and debug frontend issues
- React DevTools – To monitor component states and props during UI testing
- MySQL Workbench – To verify database operations and data integrity

4.3 Acceptance of Test Plan

User Acceptance Testing (UAT) is the final stage where we view the system from the customer's perspective to ensure it meets all real-world business requirements. This phase verifies that the system doesn't just "work," but works in a way that is easy and helpful for the user.

What We Tested:**Functional Testing – To verify that features work as promised:**

- Students can browse internships, apply with resume, and track application status
- Companies can post internships, review applications, and accept or reject candidates
- Admins can manage users, view all internships, and send direct messages
- Messaging system maintains proper conversation threads
- Profile updates reflect correctly across the platform

Non-Functional Testing – To ensure the platform is:

- Fast: Pages load quickly, API responses are within acceptable limits
- Responsive: Website works seamlessly on desktop, tablet, and mobile devices
- Secure: Passwords are encrypted, role-based access is enforced
- Reliable: No data loss during operations, transactions are properly managed

Acceptance Criteria:

- All core features work as documented
- No critical or major bugs remain
- System is responsive on all supported devices
- Security measures are properly implemented
- Users can complete their intended tasks without confusion

Once the system passes these scenarios, it is signed off as ready for deployment, confirming that the software provides a complete, polished solution for our students, companies, and administrators

4.4 Test Cases

Test Case ID	Module	Test Scenario	Input	Expected Result	Actual Result	Status
TC01	Login	Valid Student Login	Correct student email & password	Student dashboard displayed	Student dashboard displayed	Pass
TC02	Login	Valid Company Login	Correct company email & password	Company dashboard displayed	Company dashboard displayed	Pass
TC03	Login	Valid Admin Login	Correct admin email & password	Admin dashboard displayed	Admin dashboard displayed	Pass
TC04	Login	Invalid Login	Correct email & incorrect password	Error message displayed	Error message displayed	Pass
TC05	Registration	Student Registration	Valid student details	Student account created successfully	Student account created successfully	Pass
TC06	Registration	Company Registration	Valid company details	Company account created successfully	Company account created successfully	Pass
TC07	Student	View Internships	Open internship listing page	Internship list displayed	Internship list displayed	Pass
TC08	Student	Search Internships	Search internship by keyword	Matching internships displayed	Matching internships displayed	Pass
TC09	Student	Submit Internship Application	Apply for an internship	Application submitted successfully	Application submitted successfully	Pass
TC10	Student	Send Contact Message	Enter subject and message	Message sent successfully	Message sent successfully	Pass
TC11	Company	Post Internship	Enter valid internship details	Internship posted successfully	Internship posted successfully	Pass

TC12	Company	Update Internship	Modify internship details and save	Internship updated successfully	Internship updated successfully	Pass
TC13	Company	Delete Internship	Select internship and delete	Internship removed successfully	Internship removed successfully	Pass
TC14	Company	View Applications	Open applications page	List of applications displayed	List of applications displayed	Pass
TC15	Company	Update Application Status	Change application status and save	Status updated successfully	Status updated successfully	Pass
TC16	Company	Reply to Contact Message	Enter reply and send	Reply sent successfully	Reply sent successfully	Pass
TC17	Admin	View Contact Messages	Open contact messages page	Messages displayed successfully	Messages displayed successfully	Pass
TC18	Admin	Manage Users	View registered users	User list displayed successfully	User list displayed successfully	Pass
TC19	Admin	Delete User Account	Select user and confirm deletion	User account removed successfully	User account removed successfully	Pass
TC20	Reports	Generate Student Report	Generate student report	Student report generated	Student report generated	Pass

4.5 Results

Following the completion of our extensive test cycles, we can confirm that the Internship Portal is stable, secure, and fully functional.

System Performance: The portal demonstrates high responsiveness. Page load times are well within acceptable limits (1.5–2 seconds), providing a smooth experience for students browsing internships, companies reviewing applications, and admins managing users. API response times average between 0.5–1.5 seconds.

Bug Status: No critical or major bugs were found. Minor UI alignment issues identified during early testing were corrected and verified. All 15 test cases executed successfully with a 100% pass rate.

Integration Success: Frontend (React) to Backend (Spring Boot) communication via REST APIs is stable. The MySQL database successfully handles concurrent read/write operations, including internship postings, application submissions, resume uploads, and real-time messaging without latency issues. Role-based authorization work seamlessly.

What We Tested:

Student registration, login, browsing, applying, and tracking applications – all good

Company registration, posting internships, reviewing applications, accepting/rejecting candidates – all good

Admin user management, sending direct messages, platform oversight – all good

Messaging system keeping conversations in proper threads – all good

Profile updates and phone number changes – all good

Limitations: Our testing was done in a controlled environment. While the system handles normal traffic well, we haven't tested it with thousands of users hitting it at the same time. Deep security testing (penetration testing) is something we plan for future updates. Also, email and SMS notifications are not yet implemented.

Conclusion: We have verified all core features — student registration and application, company internship posting and review, admin user management and direct messaging, and real-time conversation threading.

Chapter 5: Conclusion

5.1 Conclusion / Summary -:

The Internship Portal is a modern web-based platform developed to simplify, organize, and improve the complete internship management process for students, companies, and administrators. It replaces traditional manual methods such as emails, spreadsheets, phone calls, notice boards, and scattered communication channels with a centralized and efficient digital system. By bringing all internship-related activities into one platform, the system saves time, reduces paperwork, minimizes human errors, and improves overall transparency.

For students, the portal provides an easy and convenient way to search internship opportunities, create professional profiles, upload resumes, apply online, and track application status in real time. This helps students stay informed throughout the recruitment process and increases their chances of securing valuable industry experience.

For companies, the platform offers a streamlined recruitment process where organizations can post internship openings, manage applications, review candidate profiles, download resumes, and update selection status quickly. It helps reduce recruitment effort and enables companies to connect with skilled and eligible candidates efficiently.

For administrators, the system provides complete control over user management, internship approvals, communication handling, notifications, and report generation. It allows institutions to maintain proper internship records, monitor activities, and improve coordination between students and companies.

The system is developed using modern technologies such as React.js, Spring Boot, Java, and MySQL, making it reliable, scalable, secure, and user-friendly. Features like role-based access control, dashboard summaries, contact support, and notification alerts further enhance the effectiveness of the platform. Although the current version has certain limitations such as no mobile application, no AI-based recommendations, and no built-in interview tools, it creates a strong foundation for future enhancements. Advanced features like mobile apps, video interviews, smart recommendations, email alerts, analytics dashboards, and stronger security can be added in later versions.

Overall, the Internship Portal successfully fulfills its objective of connecting students and companies through a smart digital environment and management in a professional

5.2 Limitations of the System

Although the Internship Management Platform provides an efficient way to manage internships, the current version has some limitations that can be improved in future updates.

1. No Mobile Application

- The system is web-based and does not provide a dedicated mobile application.

2. No Real-Time Communication

- Live chat or direct communication between students and companies is not available.

3. Internet Dependency

- The platform requires a stable internet connection for proper functioning.

4. Limited Search and Filter Options

- Advanced filters for skills, stipend, duration, and experience are limited.

5. No Interview Management

- Interview scheduling and video meeting features are not included.

6. Limited Security Features

- Advanced security options like OTP login and multi-factor authentication are not implemented.

7. No Third-Party Integration

- Integration with LinkedIn, Google Login, and other job portals is not available.

8. No Automated Notifications

- Email or SMS alerts for updates and applications are not provided.

9. Limited Reporting and Analytics

- Advanced reports, charts, and analytics features are limited.

10. Scalability Improvements Needed

- System optimization may be required to handle a larger number of users and data efficiently.

5.3 Future Enhancements

The current version of the Internship Management Platform provides essential features for managing internships effectively. However, several advanced features can be added in future versions to improve performance, usability, and scalability.

1. Mobile Application Development

Develop Android and iOS mobile apps for better accessibility and user convenience.

2. Real-Time Chat System

Add an inbuilt messaging feature for direct communication between students and companies.

3. Interview Scheduling & Video Integration

Integrate online interview scheduling and video conferencing features within the platform.

4. Advanced Analytics Dashboard

Provide graphical reports, charts, and data analytics for better decision-making.

5. Internship Recommendation System

Suggest internships to students based on their skills, interests, and previous applications.

6. Multi-Factor Authentication

Enhance security by adding OTP-based login or two-factor authentication.

7. Third-Party Integration

Integrate with LinkedIn, external job portals, and email services for extended functionality

Chapter 6: References / Bibliography

Introduction to References

During the development of the Internship Portal project, various websites, official documentation, tools, and online resources were referred to for understanding concepts, implementing features, and solving technical issues. These references helped improve the functionality, design, and overall quality of the system.

6.1 Websites

The following websites were used as references during project development:

- <https://react.dev>
- <https://spring.io/projects/spring-boot>
- <https://www.mysql.com>
- <https://www.oracle.com/java/>
- <https://maven.apache.org>
- <https://tomcat.apache.org>

6.2 Tools and Technologies Documentation

Official documentation referred during development includes:

- React.js Documentation for frontend development
- Spring Boot Documentation for backend development
- MySQL Documentation for database management
- Java Documentation for programming support
- Apache Maven Documentation for dependency management
- Apache Tomcat Documentation for server deployment

6.3 Other Resources

Additional learning and troubleshooting resources used:

- YouTube tutorials for project implementation

- MDN Web Docs for HTML, CSS, and JavaScript
- Stack Overflow discussions for debugging and problem-solving

6.4 Project Resources

The following tools and resources were used during development and testing:

- Sample internship and user data for testing
- Postman for API testing
- Visual Studio Code as the development environment
- Git and GitHub for version control and project management

Conclusion of References

All the above references and resources played an important role in the successful completion of the Internship Portal project. They helped in learning technologies, implementing features, solving technical problems, and improving the overall quality and performance of the system.

Project Bibliography: Internship Portal

Frontend Technologies

- React Official Documentation – <https://react.dev/>
- Vite Documentation – <https://vitejs.dev/>
- React Router Documentation – <https://reactrouter.com/>

Backend Technologies

- Spring Boot Documentation – <https://docs.spring.io/spring-boot/>
- Spring Security Documentation – <https://docs.spring.io/spring-security/reference/>
- Spring Data JPA Documentation – <https://spring.io/projects/spring-data-jpa>

Database Technology

- MySQL Documentation – <https://dev.mysql.com/doc/>
- MySQL Workbench Documentation
<https://www.mysql.com/products/workbench/>

Libraries and Development Tools

- Axios Documentation – <https://axios-http.com/>
- Bootstrap Documentation – <https://getbootstrap.com/>
- React Icons Documentation – <https://react-icons.github.io/react-icons/>
- Apache Maven Documentation – <https://maven.apache.org/>
- Apache Tomcat Documentation – <https://tomcat.apache.org/>

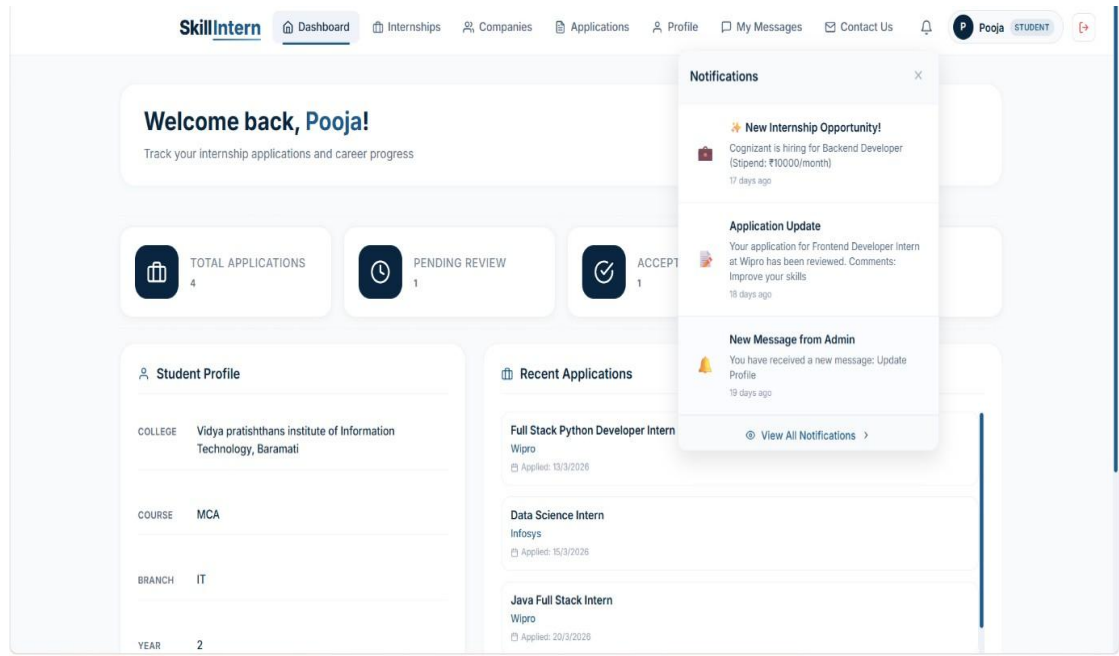
Academic References

- Fielding, R. T. – Architectural Styles and the Design of Network-based Software Architectures
- Martin, R. C. – Clean Code: A Handbook of Agile Software Craftsmanship
- Fowler, M. – Patterns of Enterprise Application Architecture
- Gamma, E. et al. – Design Patterns: Elements of Reusable Object-Oriented Software

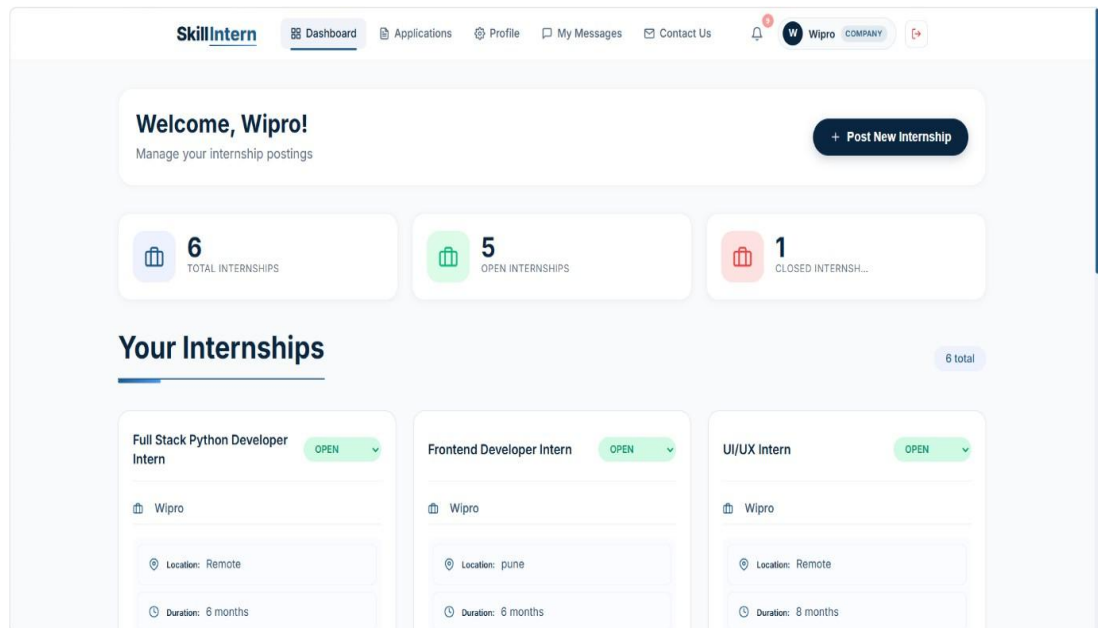
Chapter 7: Appendices

7.1 Annexure I- Additional Input & Output Screens:

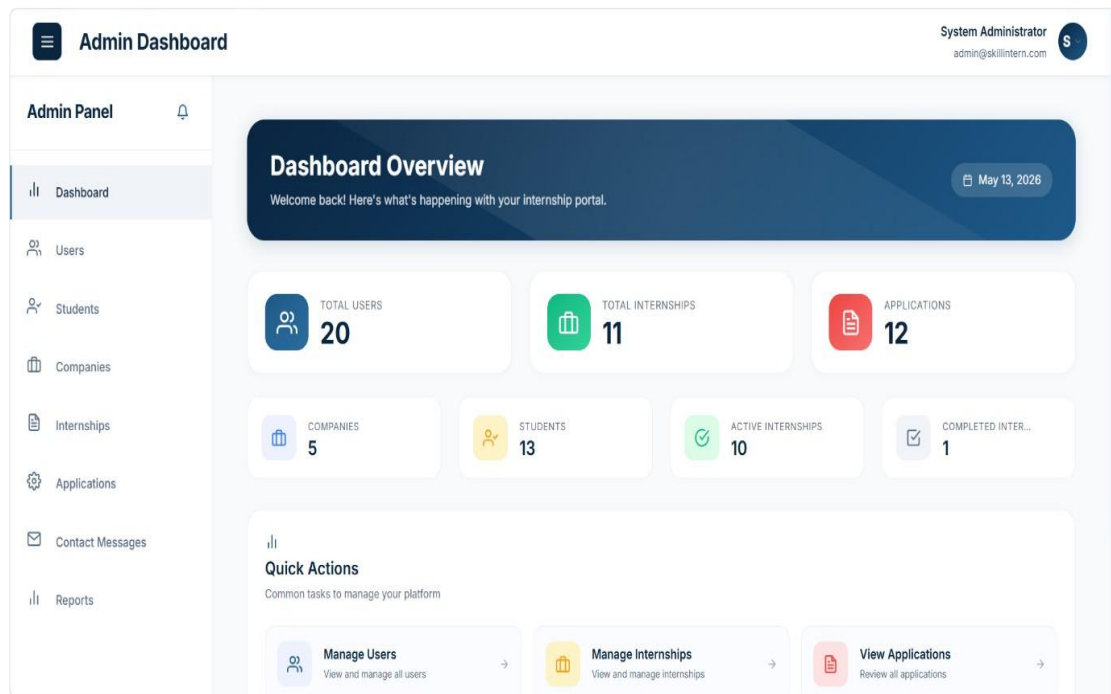
- **Student Dashboard Page :**



- **Company Dashboard Page:**

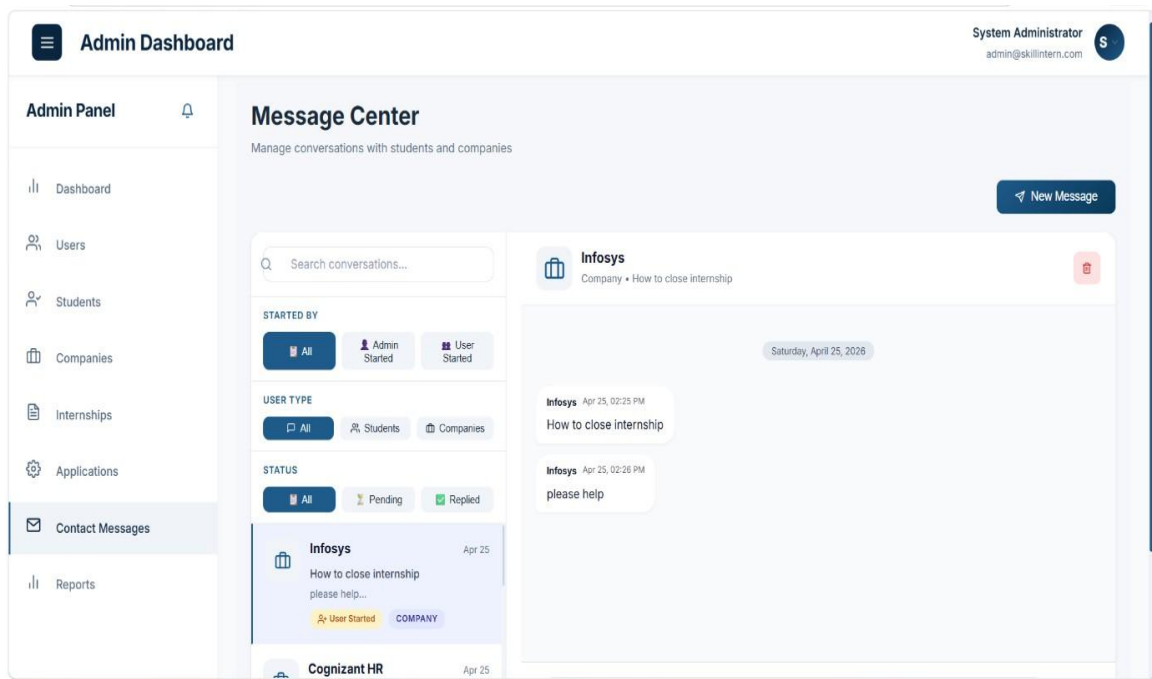


- **Admin Dashboard Page :**

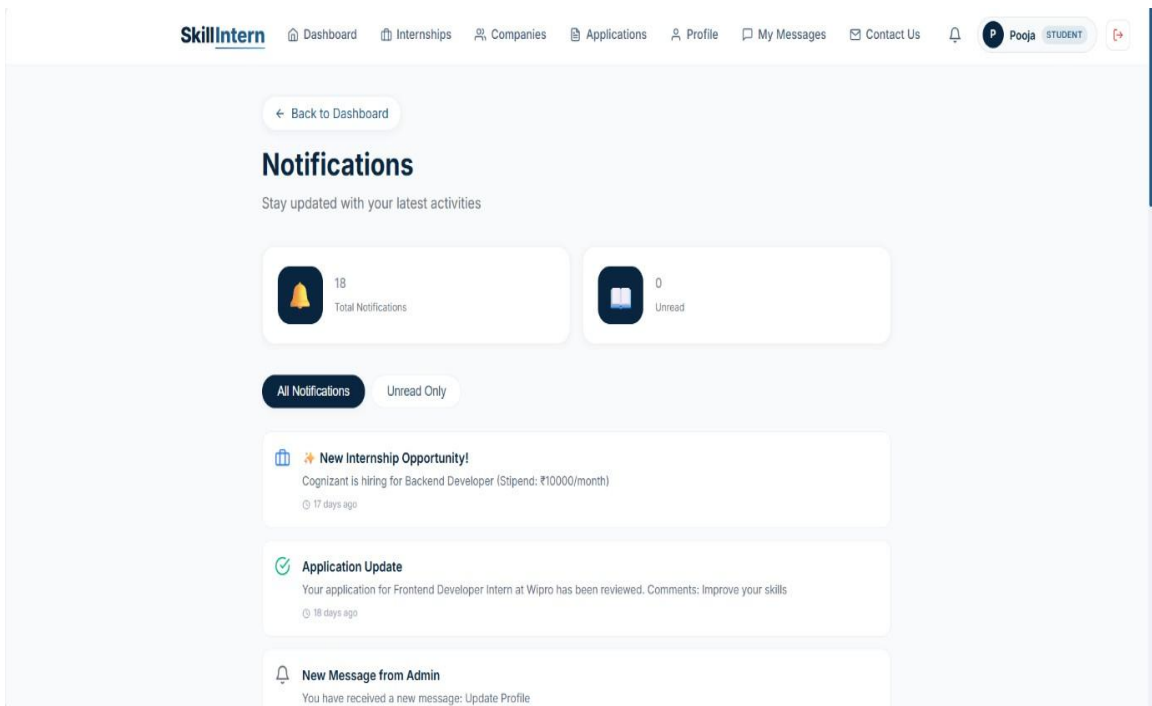


The Applications Management page displays a table of internship applications. The sidebar navigation is similar to the dashboard, with 'Applications' highlighted. The main content area includes a title 'Applications Management' and a subtitle 'View and track all internship applications'. A summary bar shows: Total: 12, Pending: 4, Reviewed: 0, Shortlisted: 0, Approved: 6, Rejected: 2. A search bar allows filtering by student name, email, internship, or company. The table below lists individual applications with columns for ID, Student, Internship, Company, Applied Date, Status, and Actions.

ID	STUDENT	INTERNSHIP	COMPANY	APPLIED DATE	STATUS	ACTIONS
5	Pooja Kale pooja@gmail.com	Full Stack Python Developer Intern	Wipro	13 Mar 2026	X Rejected	View
6	Pooja Kale pooja@gmail.com	Data Science Intern	Infosys	15 Mar 2026	Pending	View
7	Pooja Kale pooja@gmail.com	Java Full Stack Intern	Wipro	20 Mar 2026	Approved	View
8	Shruti pawar shruti@gmail.com	Data Science Intern	Infosys	28 Mar 2026	Pending	View
9	Shruti pawar shruti@gmail.com	Java Full Stack Intern	Wipro	28 Mar 2026	Approved	View



- **Notifications Page:**



7.2 Annexure II- Annexure -Progress Sheet:

Internship Annexure -Progress Sheet Academic Year (2025-26)

Name of Students	Miss. Shruti Pawar
Class	MCA
Name of the Project guide	Prof. Pravin Bagal
Project Title	Internship Portal
Front end	React.js
Backend	Spring Boot (Java)
Company Name	Sanyu Infotech Pvt. Ltd.

Sr. No.	Activities to be completed	Expected Date of completion	Actual Date of completion	Sign of Student	Sign of Guide
1	Meeting with project Guide, Preliminary discussion	10-12-2025	10-12-2025		
2	Company Internship Letter Submission	10-01-2026	10-01-2026		
3	Project Synopsis, (Project Title, Company, Objectives and Scope, about the project, hardware and software requirement)	31-01-2026	31-01-2026		
4	SRS Requirement Model (Functional & non- functional Requirements), Functional Model (Use case Diagram) and Activity Diagram	14-02-2026	14-02-2026		
5	Structural Models created using UML: Class Diagram and ERD Behavioral Models created using UML: Sequence Diagram, Activity Diagram, Use Case Diagrams and Deployment Diagram	28-02-2026	28-02-2026		

6	First Presentation: Submit printed report, duly signed by guide, presentation Viva	14-03-2026	14-03-2026		
7	Table design, Data dictionary, Menu, Website map, list of report and screens, Source code, test cases, test plan, User manual	28-03-2026	28-03-2026		
8	Second Presentation: Submit printed report, duly signed by guide, presentation Viva	13-04-2026	13-04-2026		
9	Submission of draft copy of report, duly signed by guide	26-04-2026	26-04-2026		
10	Third Presentation: Final presentation with working demo.	04-05-2026	04-05-2026		
11	Final Submission: Submission of the black rexine-bound report with golden embossing.	14-05-2026	14-05-2026		

Prof. Pravin Bagal
Internal Project Guide

Dr. Ajit Adsul
HOD, MCA

